Exact Wilson Blocking and the Yang–Mills Route
Formal Estimates and the Weak-Coupling Bridge

The MeTTapedia formalization effort

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Abstract

Ben Goertzel's Yang–Mills programme combines Wilson lattice gauge theory, renormalization, a stopping-scale polymer expansion, observable transport and Osterwalder–Schrader reconstruction. This report separates the mechanism from its proved estimates and its remaining analytic inputs. For the actual SU(2) ordered-product block map, Lean now verifies exact Haar integration, generated interaction functions, finite-depth composition and a volume-uniform contraction in a spatially weighted norm on an explicit strong-coupling interval. It also verifies a gap 1/4096 for the eliminated Maxwell fluctuations modulo gauge, with the Wilson quadratic normalization and the associated Gaussian covariance bound. The gap is on the discarded sector; it does not assert a mass for the full Maxwell field.

The missing connection is quantitative control of the actual weak-coupling trajectory and its transported observables. Exact covariance transport retains a conditional-covariance remainder at each scale. The available estimates do not yet give spatial decay for those remainders at arbitrary depth and weak coupling. We state the required entrance and transport conditions, the proved physical-distance consequences, and the further continuum obligations. Earlier extraction audits and numerical diagnostics are retained in appendices. No continuum Yang–Mills construction or mass-gap theorem is claimed.

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1 The mechanism and the present proof target

The proposed mechanism is to integrate out short-distance fluctuations while retaining the variables that carry long-distance physics. At a suitable physical scale, the generated interactions should enter a regime where a convergent polymer estimate proves spatial decay. Exact observable transport must then carry that decay back to the original lattice. A continuum limit and quantum-field reconstruction are further steps, not consequences of blocking alone.

The official problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ with positive mass gap for every compact simple gauge group [1]. The concrete blocking results in this report concern SU(2) in four dimensions. Some lattice clustering and transfer results apply to wider classes of groups. They do not supply the full group quantifier or the continuum construction.

Write $U_\ell \in \mathrm{SU}(2)$ for a positive oriented link variable and

$$U_{x,\mu\nu} = U_{x,\mu} U_{x+e_\mu,\nu} U^{-1}_{x+e_\nu,\mu} U^{-1}_{x,\nu}, \qquad V(U) = 1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr} U.$$

With product normalized Haar measure dH, the finite-volume Wilson law is

$$d\mu_\beta(U) = Z_\beta^{-1} W_\beta(U)\, dH(U), \qquad W_\beta(U) = \exp\!\left(-\beta \sum_p V(U_p)\right), \quad 0 \le V \le 2.$$

Small β is strong coupling; the continuum route needs a cutoff $a \to 0$ and a weak-coupling starting sequence $\beta \to \infty$.

The current dependency chain is

Wilson density and exact block integration (proved)
↓
weak-coupling entrance and spatial observable transport (open)
↓
small spatial interaction norm and clustering (proved)
↓
nontrivial continuum limit and OS reconstruction (open).

The new quadratic fluctuation gap supports the second line by ruling out arbitrarily soft physical modes in the eliminated linear sector. It is not the missing nonlinear estimate itself.

Goertzel's v14 proposes extended extraction through canonical dimension 16 to make irrelevant interactions contract despite recombination losses. That proposal motivates the finite coordinate audits in the appendices. The contraction proved below instead acts on actual generated interaction functions in a specified strong-coupling norm. It does not validate v14's advertised weak-coupling recursion constant.

2 Exact blocking remembers the measure and the observables

On a fine torus of side $2L$, $L = n + 3 \ge 3$, the coarse links are

$$B(U)(x,\mu) = U(2x,\mu) U(2x + e_\mu, \mu). \tag{1}$$

The order of multiplication matters. A Haar coordinate chart writes each two-link pair as $(f, f^{-1}c)$, where c is the retained coarse link; the other fine links are additional eliminated coordinates. The

chart is measure preserving, and the block map is gauge equivariant with coarse gauge transformation $g(2x)$.

For an actual positive continuous density W, define

$$\overline{W}(c) = \int W(\mathrm{chart}(c,f))\, dH(f), \qquad PF(c) = \frac{\int W(\mathrm{chart}(c,f)) F(\mathrm{chart}(c,f))\, dH(f)}{\overline{W}(c)}. \tag{2}$$

Positivity supplies the denominator. Partition functions and expectations agree before and after integration. For every finite nested elimination sequence, the direct integral agrees with staged integration, both for the density and for transported observables. The generated measure is always $\overline{W}\, dH / \int \overline{W}$; it is not replaced by unweighted Haar. For j scale changes, the fine side is $2^j(n+3)$ and the final side is $n+3$. This is an actual change of lattice scale, in addition to marginalization.

The essential two-point identity is

$$\mathrm{Cov}_{\mu_W}(F,G) = \mathrm{Cov}_{\mu_{\overline{W}}}(PF, PG) + \mathbb{E}_{\mu_{\overline{W}}}[P(FG) - (PF)(PG)]\,. \tag{3}$$

The last term records correlations carried by the discarded variables. It vanishes for exact pullbacks of coarse observables. For general original observables, iteration gives the final coarse covariance plus a sum of scale contributions $c_i(F,G)$ under the successive generated laws. Lean proves

$$\sum_{i=0}^{j-1} |c_i(F,G)| \le \tfrac{1}{2}\left(\mathrm{Var}_{\mu_W} F + \mathrm{Var}_{\mu_W} G\right). \tag{4}$$

This bound has no depth factor, but also no spatial decay. For centered Wilson plaquette marks with explicit normalization factors z, w, the right-hand side is at most $(z^2+w^2)/2$, for every real β. A cutoff-dependent normalization must therefore stay in the estimate.

There is a formal reason to keep the remainder. A positive gauge-invariant family of the form $W = 1 + \varepsilon FG$, with separated Haar-centered plaquette marks, can have $\overline{W} = 1$ and $PF = PG = 0$ while its original covariance is nonzero. This is a counterexample to inferring fine correlations from the coarse density and means alone. It is not a counterexample to the Wilson trajectory.

The GSLT route encodes these actual integration steps and expectation identities. The OSLF predicates express analytic contracts on the resulting weighted states. For example, the forward existential modality is discharged by the constructed blocked state and its proved bound. The logical interface organizes the composition; it does not supply an analytic hypothesis that has not been proved for the generated law.

3 A repeatable spatial contraction at strong coupling

An interaction state consists of continuous local functions Φ_S indexed by finite link sets, an explicit vacuum constant v, and the density

$$W_\Phi(U) = \exp\left(v + \sum_S \Phi_S(U)\right).$$

Each Φ_S is gauge invariant, depends only on S, has Haar mean zero, and $\Phi_\varnothing = 0$. Centering transfers constants into v rather than discarding normalization. The state carries a small support-weighted norm.

Links use the periodic circle sup pseudometric induced by their base points; directions at the same base point have distance zero. Let $\tau(S)$ be the infimum of lengths of finite metric tree

networks spanning S, allowing auxiliary vertices. This is the precise tree-length convention in the Lean definition, rather than a restriction to trees whose vertex set is exactly S. Set

$$\|\Phi\|_\eta = \sup_\ell \sum_{S \ni \ell} e^{8|S|+\eta\tau(S)} \|\Phi_S\|_\infty. \tag{5}$$

The coarse support map is nonexpanding in the corresponding lattice-unit metrics and sends both links in each ordered pair to its coarse link.

Theorem 1 (Actual spatial blocking; Lean)**.** *For every $\eta \geq 0$, every volume and every interaction state with $\|\Phi\|_\eta \leq 1/2$, the exact Haar integration and centered extraction produce an interaction state* $\mathrm{RG}(\Phi)$ *such that*

$$\overline{W}_\Phi = W_{\mathrm{RG}(\Phi)}, \qquad \|\,\mathrm{RG}(\Phi)\|_\eta \leq q\|\Phi\|_\eta, \qquad q = 4e^{-2} < 1.$$

The equality includes the vacuum constant. For any finite depth j, the generated interaction norm is at most $q^j\|\Phi\|_\eta$.

The quantified object is an interaction state with the stated locality, centering, gauge and smallness properties; no arbitrary-density contraction is asserted. The generated interactions are the complete functions defined by the convergent expansion, not a fitted Wilson coupling or a finite Taylor truncation. This does not give an all-order boundary-derivative contract.

For the actual Wilson starting interaction, Lean proves

$$\|\Phi^{\mathrm{W}}_\beta\|_\eta \leq 64|\beta|e^{32+4\eta}, \qquad \beta_{\mathrm{sc}}(\eta) = \frac{1}{128e^{32+4\eta}}. \tag{6}$$

Thus $|\beta| \leq \beta_{\mathrm{sc}}(\eta)$ enters the radius $1/2$, with η, the radius, q and the threshold chosen before the volume and the depth. For physical strong coupling one restricts to $\beta \geq 0$. The retained sector is nonvacuous: ordered-product Wilson observables survive the construction.

For $\eta > 0$, the same radius supplies a spatial polymer majorant and `CrossingInput` for the generated density. For continuous real observables F, G supported on finite sets S, T, with $S \neq \varnothing$ and $|F| \leq B$, $|G| \leq C$, the proved consequence is

$$|\,\mathrm{Cov}(F, G)| \leq 3BCe^{3|S|+3|T|}e^{-\eta d(S,T)}. \tag{7}$$

It also holds for their exact fine-lattice pullbacks under any finite number of these block maps. The vacuum factor cancels in normalized expectations.

Responses and fixed-depth discarded terms. The Wilson response calculus includes spatially summable second, third and fourth score-cumulant estimates on both established assembly levels. For each permitted strong-coupling parameter, the positive rates and finite constants precede volume, pinned boundary data and rotation frames. Ordered third log derivatives retain direct action contacts and all three mixed covariance corrections; conditioning retains all three corrections in the third-cumulant tower law and the fifteen partitions in the fourth-order law. Near, overlapping and coincident supports are included. These finite-order results do not establish arbitrary-order analytic control.

Actual conditional spatial bounds are also proved for one and two dyadic eliminations, uniformly in the retained field, on explicit smaller intervals

$$|\beta| \leq \frac{1}{256e^{32+12\eta}}, \qquad |\beta| \leq \frac{1}{256e^{64+56\eta}},$$

respectively. The estimates keep the enlarged dependency supports and use the generated weighted measure at the second step. They are not yet spatial bounds for arbitrary depth at weak coupling.

Other lattice consequences. Independent compact-group results prove strong-coupling clustering, vanishing central-involution twist free energy, and a transfer-operator gap. For the latter, the group is compact Hausdorff and second countable, the Borel reference probability law is inversion invariant, and the continuous potential is a real class function with $\psi(g^{-1}) = \psi(g)$ and $0 \le \psi \le 2$. At the explicit threshold in `StrongCouplingTransferGap.Rectangular.beta0`, there is $\kappa > 0$ chosen before spatial volume such that the real L^2 transfer operator has a simple positive leading eigenvalue λ_0, and every other point λ of its real spectrum obeys $|\lambda| \le e^{-\kappa}\lambda_0$. Normalized Haar satisfies the reference law hypothesis. Positivity of the integral kernel is not a claim that every eigenvalue is nonnegative.

Site reflection positivity is proved separately on the even tori of side $2(n+2)$, with an inversion-invariant reference probability and a bounded measurable real class potential invariant under inversion. At every real β, the reflection form is real and nonnegative for bounded measurable complex observables depending on links in the closed half torus. These are lattice statements. Neither the strong-coupling gap nor site reflection positivity constructs a nontrivial continuum theory.

4 The eliminated quadratic modes have a uniform gap

At the trivial background, the derivative of the actual ordered product (1) is

$$QA(x,\mu) = A(2x,\mu) + A(2x + e_\mu, \mu). \tag{8}$$

This is proved for differentiable SU(2) matrix paths through the identity. A path in the exact identity block fiber has $QA = 0$, and the Haar chart realizes the corresponding eliminated tangent directions.

For one real Lie-algebra component, define

$$\nabla\phi(x,\mu) = \phi(x + e_\mu) - \phi(x), \qquad \mathcal{G}_n = \{\nabla\phi : Q\nabla\phi = 0\}, \qquad \mathcal{H}_n = \ker Q \cap \mathcal{G}_n^{\perp}.$$

The equality $Q\nabla\phi = \nabla(\phi \circ (2\,\cdot))$ specifies exactly which infinitesimal gauge transformations preserve the coarse field. All norms here use counting measure on fine links and positive plaquettes.

Theorem 2 (Uniform eliminated-sector gap; Lean)**.** *On every four-dimensional torus of side* $2(n+3)$ *and for every real link field* A *with* $QA = 0$*,*

$$\frac{1}{4096}\,\mathrm{dist}_2(A, \mathcal{G}_n)^2 \le \sum_p (\mathrm{curl}\, A(p))^2. \tag{9}$$

Consequently $\|A\|_2^2 \le 4096 \|\mathrm{curl}\, A\|_2^2$ *on* $\mathcal{H}_n$*. The constant precedes the volume quantifier. The componentwise bound also holds for the three real colors of* $\mathfrak{su}(2)$*.*

The proof is local: a coordinate-path gauge subtracts a fiber-preserving gradient, and a finite cell Stokes identity expresses each remaining link through nearby curls. Summation gives the displayed constant, including periodic seams. The physical sector is nonzero: an explicit transverse eliminated field has nonzero curl. Thus the theorem is neither an estimate on pure gauge fields alone nor a numerical spectral extrapolation.

For $A = i\sum_{a=1}^{3} A^a \sigma_a$ in the Pauli convention, the exact identity $V(U) = \frac{1}{4}\|U - I\|_{\mathrm{HS}}^2$ fixes the quadratic coefficient:

$$\lim_{t\to 0} \frac{\beta \sum_p V(U_p(t))}{t^2} = \frac{\beta}{2} \sum_{p,a} (\mathrm{curl}\, A^a(p))^2. \tag{10}$$

The analogous centered conditional action limit is proved on the exact identity fiber. This is an action expansion relative to Haar, not a proof that the nonlinear conditional measure converges to a Euclidean Gaussian; the coordinate Jacobian and the error estimates still matter.

For $\beta > 0$, the formalized one-color centered Gaussian on $\mathcal{H}_n$ has precision $\beta D_n^* D_n$, where D_n is curl restricted to $\mathcal{H}_n$. Its covariance is the inverse precision, and

$$\mathrm{Var}\langle u, A\rangle \le \frac{4096}{\beta}\|u\|_2^2, \qquad \mathrm{Var}(\mathrm{curl}\, A(p)) \le \frac{65536}{\beta}.$$

These are volume-independent size bounds. An off-diagonal spatial decay estimate for the constrained Gaussian curvature covariance is not yet proved. The full Maxwell field still has slow modes; the theorem concerns only fluctuations removed by this specific block map.

Numerical diagnostic, separate from the proof. The reproducible Fourier diagnostic checks the constrained physical sector after removing gauge zero modes. Across the refined set of 1460 sampled momenta, including momenta of order $2\pi/10^8$, the smallest computed physical eigenvalue was approximately 0.7084973779. Eigen-equation, gauge and covariance-inverse residuals were at most $3.72\cdot 10^{-14}$, $4.25\cdot 10^{-15}$ and $1.81\cdot 10^{-14}$, respectively. These are floating-point diagnostics, not certified error intervals or a proof of the uniform minimum. The independent Lean cell argument proves the smaller constant $1/4096$.

5 The stopping scale in physical units

Fix a cutoff label k, spacing $a_k > 0$, coupling $\beta_k \ge 0$ and finite blocking depth $j(k)$ independent of spatial volume. The coarse spacing is $r_k = 2^{j(k)} a_k$. The formal entrance obligation requires constants

$$\eta > 0, \qquad 0 < r_{\min} \le r_{\max} < \infty$$

chosen before k and the volume, with $r_{\min} \le r_k \le r_{\max}$. For every cutoff and volume, the actual directly integrated Wilson density must equal an interaction-state density with $\|\Phi_{k,n}\|_\eta \le 1/2$, including its vacuum factor. The weak-coupling obligation adds $a_k \to 0$ and $\beta_k \to \infty$. Its existence is not proved.

Under entrance, the lattice decay rate becomes

$$m = \eta/r_{\max} > 0, \qquad e^{-\eta d(S,T)} \le e^{-m r_k d(S,T)}. \tag{11}$$

For exact coarse pullbacks with support sizes at most M and amplitudes B, C, Lean obtains $3BCe^{6M}e^{-mD}$ at physical separation $D = r_k d(S,T)$, uniformly in cutoff and volume.

General original observables require more. Their transported means may be sums of local terms with large supports; the formal contract charges these through anchored spatial budgets B, C. If the physical anchor separation is at least $D - \delta$ and the expected conditional-covariance remainder is at most εe^{-mD}, then Lean proves

$$|\,\mathrm{Cov}_{\mu_{\beta_k}}(F, G)| \le \left(3BCe^{m\delta} + \varepsilon\right) e^{-mD}. \tag{12}$$

All four transport constants precede the cutoff and volume. For exact pullbacks, the remainder is zero. For general marks, its bound is a separate hypothesis, not part of entrance for the density alone.

There is also a proved scale-summation criterion. If $D \geq D_0 > 0$ and each actual scale contribution satisfies

$$|c_i(F,G)| \leq A \exp\left(-\frac{\eta D}{2^i a_k}\right), \qquad 0 \leq i < j(k), \tag{13}$$

with A fixed before cutoff, volume and depth, their sum is bounded by

$$\frac{Ae^{-mD_0}}{1-e^{-mD_0}} e^{-mD}.$$

No growing depth factor is needed. Near and coincident marks can instead use the original amplitude bound; this adds $2B_0C_0e^{mD_0}$ to the prefactor when $|F| \leq B_0$ and $|G| \leq C_0$. This does not make cutoff-growing field normalizations uniformly bounded.

6 What remains to connect weak coupling to the continuum

The exact integration and strong-coupling contraction are established. The next analytic target is (13) for the actual generated Wilson trajectory, together with spatial budgets for its transported means. At scale i, the relevant conditional law is explicitly

$$d\nu_i(f \mid c) = \frac{W_i(\mathrm{chart}(c,f))}{W_{i+1}(c)}\, dH(f), \qquad d\mu_{i+1}(c) = \frac{W_{i+1}(c)}{\int W_{i+1}\, dH}\, dH(c).$$

The scale contribution averages a conditional covariance under ν_i against μ_{i+1}, using the correctly transported original marks. This formulation permits an averaged large-field estimate; it does not require a false uniform quadratic approximation at every retained configuration.

Four distinct obligations remain.

1. **Spatial weak-coupling fluctuation control.** First obtain off-diagonal decay for the constrained Gaussian curvature covariance. Then control the non-Abelian action remainder, the Haar coordinate factor, background dependence and large fields under the actual generated conditional measures. A positive quadratic gap at the identity does not imply these bounds. Plaquette energy marks are quadratic to leading order, so fourth-moment control and their normalization also enter.

2. **A weak-to-strong entrance trajectory.** Prove that a weak-coupling Wilson sequence reaches the spatial interaction radius at a volume-independent finite depth, with physical stopping scales bounded above and below. Keep all generated non-Wilson interactions. A running scalar coupling or a fixed strong-coupling example does not establish this entrance theorem.

3. **Closure of the required response information.** The available low-order response identities and exact interaction-function expansions have different scopes. If repeated weak-coupling analysis needs an all-order response contract, its order dependence, conditioning identities and remainder estimates must be proved. The dimension-16 extraction branch also still needs its complete coupled physical quotient, conditioned dual coordinates and analytic comparison to the actual Wilson interaction norm.

4. **A nontrivial continuum quantum field theory.** Construct compatible cutoff limits with suitable observable normalization, Euclidean symmetry and regularity, reflection positivity and a nontriviality witness; then perform the required reconstruction and transfer the physical decay estimate to a Hamiltonian gap. Lattice reflection positivity and a finite-volume transfer gap are inputs to this programme, not a completed reconstruction.

The most plausible failure of the proposed bridge is loss of quantitative control of physically normalized observables under repeated weak-coupling integration, even when the final marginal density looks harmless. The exact covariance identity exposes this possibility. The Maxwell gap is encouraging because the chosen block does not leave arbitrarily soft physical fluctuations in its eliminated quadratic sector. Deciding whether the nonlinear generated measures retain useful spatial control is the substantive next test.

Calibration and extraction audits. The crossing criterion is calibrated against the free Maxwell field: `FreeFieldNoCrossing.false_of_phaseRepresentation` rules out reproducing its specified finite-torus Gaussian phase expectations by a crossing family. The twist diagnostic is a necessary condition for a twist-compatible crossing representation; vanishing twist free energy alone is not a sufficient crossing theorem. Earlier finite-volume Monte Carlo tests of bare Wilson and hierarchical models are retained in Appendix D. They do not refute the actual generated nonlinear trajectory by fitting it to a single coupling.

The extraction audit likewise distinguishes a valid scalar calculation from an unproved analytic constant. The arithmetic $2224\,2^{-13} = 139/512$ is checked, but v14's two factor ledgers give 33152 and 11088/5, and neither derives the needed Wilson recursion estimate as written. A projection's range does not control its norm. The repaired finite constructions preserve trace order and the coupling of field sectors by commutators; complete joint conditioning remains an additional task. These findings delimit particular steps of the route rather than deciding the continuum question.

7 Formal sources and evidence

The formal results below refer to the Lean source snapshot `1e8812ebd` and its dependencies. The full semantic index is the Yang–Mills module README. The following short map identifies the principal current claims.

Exact integration and spatial contraction. `SU2WilsonAssembly`, especially `InteractionState.spatial_norm_block_le`; `WilsonDyadicGeometry`; and `WilsonDyadicObservableFlow`, especially `wilson_spatial_iteration_bound`, `covariance_eq_direct_add_scaleContributions`, `sum_abs_scaleCovarianceContribution_le` and `StoppingScaleEntrance.uniform_transport_clustering`.

Conditional observables and responses. `WilsonConditionalProjection`, `WilsonBoundaryResponse`, `WilsonLocalResponse`, `WilsonProjectionComposition` and `SU2WilsonAssemblyLocality`; `MarginalBlindness` contains the discarded-covariance obstruction.

Eliminated Maxwell modes. `WilsonMaxwellBlock`, especially `hasDerivAt_blockField`, `uniform_fluctuation_gap`, `Gaussian.covariance_precision` and `Gaussian.variance_le`; the identity-fiber action and conditional normalization are integrated in `SU2WilsonAssembly`. The diagnostic is `maxwell_block_gap_diagnostic.py`.

Lattice clustering, twists and spectral statements. `CrossingInput`, `StrongCouplingCrossing`, `TwistDiagnostic`, `StrongCouplingTwist`, `WilsonReflectionPositivity`, `CompactTransferGap`, `CalibrationSeparation` and `FreeFieldNoCrossing`.

Extraction and endpoint scope. `RGBootstrap`, `ExtendedExtractionTransfer`, `WilsonExtractionRecombinationConstant` and `ProofState`; the appendices retain the named finite-coordinate, trace-order and quotient certificates.

The latest Maxwell integration was checked together with its affected consumers; an exhaustive declaration audit of the nine-module change covered 2038 declarations and found only `propext`, `Classical.choice` and `Quot.sound`. This is the scope of that audit, not a claim that every source named in the historical appendices was rebuilt in the present documentation revision. The source snapshot uses Lean 4.34.0-rc2 and Mathlib `85e3a25`. Exact finite certificates are distinguished below from analytic comparisons and floating-point experiments.

Primary literature and applicability. Dimock's covariant axial-gauge construction provides a relevant comparison for constrained Abelian fluctuation estimates [2]. Its translated path averages, axial constraints and odd blocking convention differ from the anchored scale-two product used here. The present bound is proved for our own map; no constant is imported across that difference. Balaban's four-dimensional small-field effective-action programme is a primary source for the broader analytic strategy [3]. It does not, by citation alone, discharge the nonlinear remainder, large-field or transport hypotheses of this formalization. Goertzel's v14 is the proposal audited in the appendices; the exact integration and local gap statements above are separate proved results within the same Wilson/RG programme.

References

[1] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, Clay Mathematics Institute problem description. `https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf`.

[2] J. Dimock, *Covariant Axial Gauge*, Letters in Mathematical Physics **105** (2015), 959–987. `https://arxiv.org/abs/1410.2798`.

[3] T. Balaban, *Renormalization group approach to lattice gauge field theories. I. Generation of effective actions in a small field approximation and a coupling constant renormalization in four dimensions*, Communications in Mathematical Physics **109** (1987), 249–301. `https://doi.org/10.1007/BF01215223`.

A Extended extraction in the v14 proposal

The following appendices preserve the detailed algebraic audits and earlier diagnostics. They concern the stated coordinate models, source policies and manuscript factorization. Their conditional constants are distinct from the actual strong-coupling interaction contraction in Theorem 1. Current analytic obligations are collected in §6.

A.1 The continuum-limit obstruction and extended extraction

In any RG scheme one must show that the "irrelevant" sector (operators of canonical dimension > 4) contracts at each step:

$$\|K_{j+1}^{\mathrm{irr}}\| \;\le\; \Lambda\,\|K_j^{\mathrm{irr}}\| + (\text{source terms}), \qquad \Lambda < 1.$$

The manuscript identifies a sharp obstruction in the *naive* scheme, where one extracts only the relevant and marginal operators of dimension ≤ 4. Under rescaling, a dimension-d operator scales by b^{4-d}, so the leading irrelevant operators (dimension 6) contract by $b^{-2} = 0.25$ for $b = 2$. But the RG step also recombines polymers, contributing a combinatorial constant C_1. The manuscript advertises the working bound $C_1 \le 2224$ for $b = 2$; if that value were established, the naive gain would be

$$\Lambda_{\mathrm{naive}} \;=\; 2224\,b^{-2} \;=\; 2224 \times 0.25 \;=\; 556 \;>\; 1,$$

an *expanding* map: initial smallness would grow like 556^J over $J \sim |\log a_0|$ steps and diverge as $a_0 \to 0$.

The proposed resolution is *extended extraction*. Instead of extracting only dimension ≤ 4 operators, one extracts all gauge-invariant local operators up to canonical dimension $d_{\max} = 16$ (for $b = 2$). The irrelevant sector then begins at dimension $d_{\max} + 1 = 17$ and contracts by

$$\lambda_{\mathrm{irr}} \;=\; b^{\,4-(d_{\max}+1)} \;=\; b^{\,3-d_{\max}} \;=\; 2^{-13} \;\approx\; 1.2 \times 10^{-4}.$$

With the same advertised value, the proposed contraction factor becomes

$$\boxed{\Lambda \;=\; C_1\,\lambda_{\mathrm{irr}} \;\le\; 2224 \cdot 2^{-13} \;\approx\; 0.27 \;<\; 1.}$$

The manuscript presents this as genuine contraction "with substantial slack." The audit below verifies this arithmetic only *conditional on the supplied constant*; it refutes the available derivations of that constant as written. The price is tracking a finite-dimensional vector of higher couplings $\vec{g}_j = (g_j^{(4)}, g_j^{(6)}, \ldots, g_j^{(d_{\max})})$ rather than the single gauge coupling; the manuscript argues these are perturbatively small and do not affect the universal one-loop beta-function coefficient $b_0 = 11/(3 \cdot 16\pi^2)$, which is determined entirely by local ultraviolet information.

A.2 The recombination constant and the extraction-projection norm

Appendix O proposes the factorization

$$C_1 \;\le\; C_{\mathrm{fluct}} \cdot C_{\mathrm{cumulant}} \cdot C_{\mathrm{recomb}} \cdot C_{\mathrm{extract}} \;\le\; 1.65 \times 3b^4 \times 14 \times 2,$$

which at $b = 2$ equals $11088/5 = 2217.6$, rather than exactly 2224. Appendix F.5 displays a different factor product which equals 33152 at $b = 2$; the two ledgers are inconsistent. The last Appendix-O factor $C_{\mathrm{extract}} = \|P_{\le d_{\max}}\|_{\mathrm{op}}$ is the operator norm of the extraction projection onto

the finite-dimensional space of operators of dimension $\le d_{\max}$. The manuscript bounds it (its Computation O.7(d)) by

$$\|P_{\le d_{\max}}\|_{\mathrm{op}} \;\le\; \sum_{d=0}^{d_{\max}} \binom{\#\text{indices}}{d} (r_1/r_0)^{-d} \;\le\; 1 + O\big((r_0/r_1)^{d_{\max}}\big),$$

and asserts $\|P_{\le d_{\max}}\|_{\mathrm{op}} \le 2$ for $r_1/r_0 = 1/2$ and $d_{\max} = 16$. The literal negative exponent grows, but changing only its sign does not turn the displayed multivariate sum into a one-dimensional geometric series. More fundamentally, v14 specifies P_{ext} by its range and does not give the dimension-16 invariant basis, dual jets, or conditioning estimates needed to determine its operator norm. We return to the proved realization and the transfer obstruction in §C.

B Scalar bootstrap and finite reconstruction

The formalization lives in the Lean namespace `Mettapedia.QuantumTheory.YangMills`. This appendix records the earlier four components: the contraction predicate, the knife-edge arithmetic, the finite strong-coupling gaps with their finite OS reconstruction, and the continuum-open gating. The listings preserve the earlier declarations with Unicode notation transliterated to ASCII for typesetting; the linked Lean source governs the precise statements.

B.1 The extended-extraction contraction predicate

The module `Mettapedia.QuantumTheory.YangMills.RGBootstrap` models the route's contraction obligation as a predicate over a recombination constant C, block factor b, and extraction depth $d_{\max}$. The irrelevant scaling factor is $b^{3-d_{\max}}$ (an integer power, since the useful cases have $d_{\max} > 3$), and a contraction is a nonnegative constant whose product with the irrelevant scale is below 1.

```
/-- Irrelevant scaling factor ‘b^(3-dmax)‘ used by the extended-extraction
discussion.  The exponent is an integer because the useful cases have
‘dmax > 3‘. -/
def irrelevantScale (b dmax : Nat) : Real :=
  (b : Real) ^ (3 - (dmax : Int))

/-- The route-level contraction audit for a combinatorial constant and an
irrelevant scaling factor. -/
def HasRGContraction (C scale : Real) : Prop :=
  0 <= C /\ 0 <= scale /\ C * scale < 1

/-- The extended-extraction audit specialized to ‘lambda = b^(3-dmax)‘. -/
def HasExtendedExtractionContraction (C : Real) (b dmax : Nat) : Prop :=
  HasRGContraction C (irrelevantScale b dmax)
```

A threshold lemma in the same module records the exact content of the predicate: for $b > 0$ and $d_{\max} \ge 3$, the contraction `HasExtendedExtractionContraction C b dmax` holds if and only if $0 \le C < b^{d_{\max}-3}$. Thus the entire arithmetic content of the route is the comparison of C with the threshold $b^{d_{\max}-3}$.

B.2 The knife-edge arithmetic

The module machine-checks that the advertised constant contracts at $d_{\max} = 16$ and that the exact one-step gain is $2224 \cdot 2^{-13} = 139/512 < 1/3$:

```
/-- The advertised numerical slack: ‘2224 * 2^-13 < 1‘. -/
theorem rgContraction_2224_two_sixteen :
    HasExtendedExtractionContraction 2224 2 16 := by
  refine <by norm_num, by rw [irrelevantScale_two_sixteen]; norm_num, ?_>
  rw [irrelevantScale_two_sixteen]
  norm_num

/-- Exact advertised one-step gain for the extended-extraction constants. -/
theorem rgGain_2224_two_sixteen_eq :
    (2224 : Real) * irrelevantScale 2 16 = (139 : Real) / 512 := by
  rw [irrelevantScale_two_sixteen]
  norm_num
```

Crucially, the same constant does *not* contract at the nearest even extraction depth below the advertised one. At $d_{\max} = 14$ the threshold is $2^{11} = 2048 < 2224$, so the contraction is arithmetically false; the module proves that any contraction at $d_{\max} = 14$ would force $C < 2048$, and hence that 2224 fails:

```
/-- Any contraction proof at ‘dmax = 14‘ must improve the linear recombination
constant below ‘2048‘.  The manuscript’s advertised bound ‘2224‘ is therefore
insufficient for this lower extraction depth. -/
theorem rgContraction_two_fourteen_forces_constant_lt_2048
    {C : Real} (h : HasExtendedExtractionContraction C 2 14) :
    C < 2048 := by
  have hprod : C * irrelevantScale 2 14 < 1 := h.2.2
  rw [irrelevantScale_two_fourteen] at hprod
  nlinarith

/-- Same-constant blocker at the nearest even extraction depth below ‘16‘. -/
theorem not_rgContraction_2224_two_fourteen :
    not  HasExtendedExtractionContraction 2224 2 14 := by
  intro h
  have hC : (2224 : Real) < 2048 :=
    rgContraction_two_fourteen_forces_constant_lt_2048 h
  norm_num at hC
```

This is the knife edge: the choice $d_{\max} = 16$ is exactly what makes the advertised constant contract, and the manuscript's own table shows $d_{\max} \in \{10, 12, 14\}$ all fail against $C_1 \leq 2224$ while $d_{\max} = 16$ succeeds. The module `Mettapedia.QuantumTheory.YangMills.RGCrux` packages this into a route-level statement: a mass-gap route that keeps the advertised constant 2224 but uses an even extraction depth strictly below 16 cannot exist, since the required contraction is arithmetically false before any OS / clustering layer is reached.

```
/-- Same-constant lower-extraction certificate in the manuscript’s even
canonical-dimension shape: the cutoff is strictly below ‘16‘, is even, and uses
the same advertised recombination bound ‘2224‘. -/
def SameConstantEvenBelowSixteenRGCertificate (dmax : Nat) : Prop :=
  dmax < 16 /\ Even dmax /\ HasExtendedExtractionContraction 2224 2 dmax

/-- No same-constant RG certificate exists for an even manuscript-shape cutoff
strictly below ‘16‘. -/
theorem not_sameConstantEvenBelowSixteenRGCertificate
```

```
    {dmax : Nat} :
    not  SameConstantEvenBelowSixteenRGCertificate dmax := by
  intro hcert
  have hdmax : dmax <= 14 :=
    even_lt_sixteen_le_fourteen hcert.1 hcert.2.1
  exact (not_rgContraction_2224_two_of_dmax_le_fourteen hdmax) hcert.2.2
```

A strengthened companion theorem, `not_atLeast2048EvenBelowSixteenRGCertificate`, shows that any even lower-extraction repair below $d_{\max} = 16$ would need a genuinely sharper constant ($C < 2048$), not merely the advertised one. We stress that this is an honest delimiter of the arithmetic surface, not a refutation of the route: at the raw natural number $d_{\max} = 15$ the same constant *does* contract ($2^{12} = 4096 > 2224$), and the module records this explicitly so that the obstruction is read precisely as an even canonical-dimension statement matching the manuscript's dimensions $4, 6, \ldots, d_{\max}$.

B.3 Finite strong-coupling spectral gaps and finite OS reconstruction

The module `Mettapedia.QuantumTheory.YangMills.FiniteOSReconstruction` formalizes the finite algebraic core of OS reconstruction: a reflection-positive finite kernel induces a positive pairing on observables, and a transfer operator that is self-adjoint for that pairing descends to the pairing quotient and remains self-adjoint there. On this base, the modules `Z2StrongCouplingGap` and `Z3StrongCouplingGap` build explicit finite single-link abelian lattice gauge models. For $\mathbb{Z}_2$, the one-link heat-kernel transfer at heat time 1 has eigenvalues 1 and $\exp(-1)$, so the dimensionless gap is $-\log(\exp(-1)/1) = 1$:

```
theorem z2StrongCouplingGap_eq_one :
    z2StrongCouplingGap = 1 := by
  simp [z2StrongCouplingGap, z2StrongCouplingLambdaTwo,
    z2StrongCouplingLambdaOne, z2StrongCouplingRatio, Real.log_exp]

/-- End-to-end finite lattice theorem: the explicit ‘Z2‘ strong-coupling model is
nontrivial, reflection positive, has an OS transfer operator, and has
dimensionless transfer gap at least ‘1‘. -/
theorem z2StrongCoupling_latticeYangMills_massGap_landmark :
    exists _ : Z2StrongCouplingGapCanary,
      (exists x y : Z2OneLinkConfig, x != y) /\
        OSReflectionPositive z2StrongCouplingKernel /\
        z2StrongCouplingGap = 1 /\
        (1 : Real) <= z2StrongCouplingGap /\
        0 < z2StrongCouplingGap := by
  exact
    <z2StrongCouplingPositiveCanary,
      z2OneLinkConfig_nontrivial,
      z2StrongCouplingKernel_reflectionPositive,
      z2StrongCouplingGap_eq_one,
      z2StrongCouplingGap_lower_bound_one,
      z2StrongCouplingGap_positive>
```

The $\mathbb{Z}_3$ module proves the analogous landmark with two mean-zero transfer eigenvectors and the same unit gap. These theorems are unconditional and non-vacuous: they exhibit genuine reflection positivity, a genuine finite OS transfer reconstruction, and a strictly positive spectral gap. They are also kept rigorously separate from any continuum claim. Each module carries an explicit scaling

diagnostic showing that a finite gap need not survive a naive continuum schedule. For the heat-time schedule $t(a) = a^2$, the *physical* gap (dimensionless gap divided by lattice spacing) closes as $a \to 0$:

```
/-- A concrete gap-closing schedule for the finite model under the heat-time
scaling `t(a)=a^2`: the physical gap can be made smaller than any positive
threshold by taking the lattice spacing small enough. -/
theorem z2QuadraticHeatTime_physicalGap_closes :
    forall eps : Real, 0 < eps ->
      exists a : Real,
        0 < a /\ a < eps /\
          z2HeatTimePhysicalGap a (a ^ 2) < eps := by
  intro eps heps
  refine <eps / 2, by linarith, by linarith, ?_>
  have hne : eps / 2 != 0 := by linarith
  rw [z2QuadraticHeatTime_physicalGap_eq hne]
  linarith
```

B.4 The continuum endpoint is gated open

The module `Mettapedia.QuantumTheory.YangMills.ProofState` records a status ledger over the route's nodes. The finite RG / extraction surfaces are marked `checked` or `refuted` as appropriate, but the continuum mass-gap endpoint carries an explicit `openGoal` status, gated behind a constructive-QFT promotion barrier (continuum measure, Euclidean covariance, reflection positivity, OS reconstruction, and Hamiltonian mass-gap transfer — none of which is represented for the RG / extraction route, which supplies only a finite ledger):

```
/-- The final mass-gap endpoint remains an open goal in this lane. -/
def yangMillsMassGapEndpointNode : YangMillsProofNode where
  id := "yang-mills.mass-gap-endpoint"
  status := .openGoal
  truthValue := <0, 95>
  evidence := "currentYangMillsMassGapEndpoint_blockedByConstructiveGate blocks promotion
     from the audited RG/extraction surfaces to a continuum mass-gap endpoint ..."
  nextObligation := "Connect the RG and extraction surfaces to a genuine continuum mass-
    gap statement only after the constructive-QFT gate, OS/Hamiltonian transfer
    dependency packet, challenge-world interface packet, and mass-gap promotion gate
    clear."

theorem yangMillsMassGapEndpointNode_open :
    yangMillsMassGapEndpointNode.status = .openGoal := by
  rfl
```

A companion theorem `currentYangMillsMassGapEndpoint_blockedByConstructiveGate` records that the endpoint is open, the constructive checks are all unrepresented, and the endpoint interface is the finite RG / extraction ledger. The newer blocking and transport theorems do not discharge this constructive gate.

C Finite extraction and quotient audits

This appendix specifies the extraction models, relation policies and finite certificates underlying the v14 audit.

The contraction constant is an input, not a conclusion. Every arithmetic theorem above takes the recombination constant as a *given real number*. The predicate `HasExtendedExtractionContraction C b dmax` is a statement about C, b, $d_{\max}$ alone; the checked facts `rgContraction_2224_two_sixteen` and `not_rgContraction_2224_two_fourteen` verify the comparison $C \lessgtr b^{d_{\max}-3}$ for the specific value $C = 2224$. Nothing in the formalization derives that 2224 (or any $O(1) \cdot b^4$ bound) is the correct value of the manuscript's recombination constant. The Lean ledger is therefore faithful to the route's bookkeeping but agnostic about the route's hardest analytic claim.

The extraction projection: realization succeeds, range-only transfer fails. The sign-only repair is insufficient: even with positive exponent, the displayed binomial/multivariate multiplicities make the corrected finite sum exceed 2 once at least two coordinates are present. We therefore constructed the natural finite polynomial realization rather than treating the omission as a verdict. For a finite Wilson block of actual group-valued links, plaquette holonomies, and a conjugation-invariant Wilson potential, finite polynomials in the plaquette coordinates are gauge invariant. In the explicitly defined weighted coefficient-majorant norm, canonical-dimension truncation has operator constant 1 (module `WilsonBlockMajorant`).

This is our realization of the manuscript's extraction idea, not an attribution to v14. It cannot be transferred to v14's P_{ext} from the specified range alone: `projectionRangeSpecification_does_not_imply_bound_two` constructs an idempotent skew projection onto the same low-coordinate range that violates the bound 2. The positive transfer theorem `coefficientL1_basisJetProjection_le` shows exactly what would suffice: explicit basis and dual-jet matrices whose coefficient-column bounds multiply. The construction attempt is certified in `Dimension16WilsonExtractionCertificate`: F.4's on-shell relation policy and O.9's off-shell policy disagree already on the displayed dimension-6 divergence-square operator; the exact block quotient, lattice representatives, finite basis, dimension-16 dual jets, biorthogonality, and conditioning matrices are not supplied. Appendix I's explicit jets concern the separate observable projection $P^{\mathrm{obs}}_{\leq 4}$. Hence the extraction component is not a standing erratum repair, and C_1 is not derived.

Hard truncation does not descend to the actual trace-word quotient. The third round replaces schematic carriers by the exact object: the SU(2), $b = 2$ Wilson block with normalized Haar measure — 81 vertices, 216 positive links of which 8 are internal, 216 plaquettes, an internal spanning tree with 7 co-tree links, exact boundary-plus-co-tree coordinates, the gauge action, the normalized Wilson potential, and the fixed-boundary product-Haar reference integral with its logarithmic effective block action (modules `SU2WilsonBlockPilot`, `SU2MajorantRepairDecision`). On this object the question "is canonical-dimension truncation even well-defined on the gauge-invariant quotient?" has a negative, intrinsic answer. For any $U \in \mathrm{SU}(2)$ the Cayley–Hamilton identity gives $\mathrm{Tr}(U^2) = \mathrm{Tr}(U)^2 - 2$; writing $a = 2 - \mathrm{Tr}(U)$ and $b = 2 - \mathrm{Tr}(U^2)$ (each of canonical dimension 4), the exact relation $b = 4a - a^2$ yields

$$a^3 b - 4a^4 + a^5 \;=\; 0,$$

whose three monomials have construction-derived canonical dimensions 16, 16, 20. Truncation through dimension 16 deletes only a^5; on the explicit holonomy $\mathrm{diag}(i, -i)$ the truncated expression evaluates to -32, while the exact left-hand side vanishes on every field. Hence the semantic kernel is not stable under representative-wise hard truncation, no function on the evaluated trace-word quotient agrees with that truncation, and the corresponding descended extraction certificate is uninhabited (module `SU2CayleyHamiltonTruncationObstruction`). The witness uses only the

Cayley–Hamilton relation — no equations-of-motion quotient and no boundary-extension convention — so it applies to both of v14's relation policies, and it refutes any uniform-in-N architecture that requires representative-wise hard truncation of redundant Wilson trace words.

Corrected constructions do carry norm-below-two extractors. Truncation is, of course, definable on a *chosen basis*, and the third round constructs the two natural repairs exactly. The full associated-graded jet chart on the gauge-fixed block (7 co-tree links, 3 Lie coordinates each, hence 21 coordinates) carries the symmetric-monomial basis of jets through total degree 16, with identity jet matrix and biorthogonal coefficient duals; radial Taylor extraction at radius ratio 1/2 has the exact operator constant $\sum_{d=0}^{16} 2^{-d} = 131071/65536 < 2$ (modules `SU2AssociatedGradedExtraction`, `SU2AssociatedGradedExtractionDecision`). Independently, the evaluated one-plaquette trace algebra is exactly a one-variable polynomial algebra: the normal-form theorem $b = 4a - a^2$ (proved semantically complete via an infinite Cayley-rotation family) gives the basis $1, a, a^2, a^3, a^4$ through dimension 16, with extraction constant $\sum_{d=0}^{4} 2^{-d} = 31/16 < 2$ (module `SU2IndependentTraceBasis`). Keeping Appendix O's other factors unchanged, the associated-graded constant yields the conditional ledger value

$$C_{\text{proposed}} \;=\; \tfrac{90832203}{40960} \;\approx\; 2217.583 \;<\; 2224,$$

whose two-source gain at depth 16 is $90832203/335544320 \approx 0.2707 < 1/3$, while depth 15 does not close. This is conditional bookkeeping: the fluctuation, cumulant, and recombination factors in this particular weak-coupling factorization remain unproved on the actual map. Two exact theorems keep these constructions honestly separated from v14's object: the full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any coordinate space of dimension at most 1851, so the manuscript's schematic count requires a genuine invariant quotient that v14 does not construct; and the radial norm used here is a precise repair, not provably the manuscript's intended multivariate norm.

Neither survivor transfers to the canonical (F, D) space. The transfer question is then decided negatively by two independent machine-checked obstructions (module `V14FDQuotientTransferNoGo`, headline `v14_fullFD_quotientTransfer_noGo`). *Derivative blindness:* v14 counts canonical dimension as $2 \cdot \#F + \#D$ (its O.9.1), but radial degree on either chart is blind to derivative factors. The scalar descendant $(\partial^2)^3[(\mathrm{Tr}F^2)^3]$ has canonical dimension 18 yet radial degree 6 on the associated-graded chart and 3 on the trace chart, so both norm-below-two extractors *fix* it — and an explicit two-cell zero-boundary sixth-difference witness evaluates to -6, so the retained operator is nonzero. Both survivors therefore fail to implement the dimension-≤ 16 projection of F.4.2, independently of the choice of quotient basis (`not_associatedRadialImplementsCanonicalCutoff`, `not_independentRadialImplementsCanonicalCutoff`). *Integration-by-parts/block incompatibility:* v14's local norm (F.3) lives on ordinary functions on a finite block retaining boundary data, while both relation branches identify integrated total derivatives with zero. The dimension-6 descendant $\partial^2\mathrm{Tr}(F^2)$ is gauge invariant and evaluates to 1 on an explicit three-plaquette block field (a purely quadratic associated-grade witness gives the same value, so higher Taylor terms are not the cause). An ordinary block-functional realization would have to send one quotient class to both the zero function and a nonzero boundary descendant; the faithful transfer sockets — exact-Wilson, pure associated-graded, and every RG-compatible extension — are therefore uninhabited under both relation policies. Consequently no corrected recursion constant propagates from the constructions above into the bootstrap. The obstruction is structural rather than informational: supplying the omitted dimension-16 coordinates or conditioning matrices cannot make either extractor into the canonical (F, D) projection. A viable replacement needs

a derivative-aware filtration whose truncation provably agrees with $2 \cdot \#F + \#D$, together with a boundary-aware complex that retains total-derivative descendants on each block as boundary cochains and imposes integration by parts only at global recombination; constructing exactly that replacement is the current object of this formalization.

The actual-lattice census cannot be Lorentz-only. The derivative-aware boundary-cochain repair makes the filtration and local boundary bookkeeping explicit, but a further obstruction appears before its coordinate basis or dual jets can be supplied. Appendix O.9.3 enumerates only Lorentz-invariant complete contractions. The exact finite-spacing Wilson lattice has only hypercubic symmetry, and the module `SU2LatticeFDCensusNoGo` supplies a gauge-invariant canonical-dimension 8 observable whose fourth Cartan jet is

$$6(F_{01}^4 + F_{02}^4 + F_{03}^4 + F_{12}^4 + F_{13}^4 + F_{23}^4).$$

An explicit rational orthogonal rotation changes its normalized quartic value from 1 to 337/625. Hence no finite span of the $O(4)$-invariant O.9.3 jets contains it. The necessary spanning core, the O.9.2 off-shell certificate, and the F.4.3 on-shell certificate are all formally uninhabited. The witness is derivative-free, so changing EOM, Bianchi, integration-by-parts, trace, Cayley–Hamilton, or boundary-cochain reduction policies cannot repair this same-target symmetry mismatch. The result is already an $N = 2$ counterexample to an architecture asserted uniformly for $\mathrm{SU}(N)$. It is not a no-go for a hypercubic-complete basis or for a controlled continuum projection.

The first hypercubic census succeeds at quartic order and refutes its padded extension. The unpadded carrier of degree-four monomials in the six oriented two-form components has cardinality 126. Exact signed-orbit certificates divide it into four admissible $H(4)$ orbit types — a pure fourth power, adjacent squares, opposite squares, and a rectangle product — plus monomials killed by negative stabilizers. Four signed orbit indicators obey the full hypercubic covariance law, their representative evaluations form the 4×4 identity, and they reconstruct every invariant rational coefficient function. This is a complete result for the derivative-free Cartan quartic restriction, not for the noncommutative (F, D) quotient. The actual gauge-invariant Wilson observable has exact fourth derivative equal to the polynomial realization of coordinate vector $(6, 0, 0, 0)$. Moreover, the curvature observables invariant under every orthogonal spacetime matrix form a proper linear subspace, and this pure-fourth polynomial defines a nonzero class in the quotient by that subspace. Thus the Lorentz-breaking artifact is a coordinate-level theorem, not merely an external rotation counterexample.

The fixed-width full carrier cannot supply that extension faithfully. Inactive field and derivative slots are mathematically padding but remain part of raw equality. The module `V14HypercubicFDCensusPaddingNoGo` constructs two distinct encodings of the empty active syntax and proves `faithful_dimension16_exactCensus_uninhabited`: if every basis column ignores inactive padding, the two basis rows coincide; the required right inverse instead makes them two different rows of the identity matrix. This obstruction precedes Bianchi, EOM, integration-by-parts, trace, and Cayley–Hamilton reduction. It does not show that an $H(4)$ basis cannot exist. It shows that the next construction must first replace the raw carrier by an unpadded/dependent syntax or quotient padding explicitly.

Our unpadded raw character census is exact through dimension 16. Formal repair. The replacement uses dependent active field and derivative slots, so no inactive coordinate enters

equality. An explicit equivalence identifies its canonical-dimension-16 subtype with a separated presentation consisting of an active trace/derivative layout and a rank-16 spacetime-axis assignment; it commutes with the literal hypercubic action and preserves the product of all active tensor signs. The induced linear equivalence identifies the scalar and pseudoscalar covariance submodules on the original syntax with the counted invariant fibers. The exact layout multiplicity is 5,361,612,271,200, and the complete raw carrier has 23,027,949,358,636,282,675,200 elements.

The standard signed four-vector representation of the literal 384-element hypercubic group is constructed over $\mathbb{Q}$. Reynolds-character averaging, with all 24×16 group elements replayed by kernel-checked certificate blocks, gives the scalar multiplicities

$$(1, 0, 1, 0, 4, 0, 31, 0, 379, 0, 5611, 0, 87979, 0, 1400491, 0, 22379179)$$

and pseudoscalar multiplicities

$$(0, 0, 0, 0, 1, 0, 20, 0, 336, 0, 5440, 0, 87296, 0, 1397760, 0, 22368256)$$

at tensor ranks 0 through 16. Since the layout coordinate is inert under $H(4)$, the rank-16 values imply exact raw invariant-space dimensions 119,988,480,745,781,344,800 and 119,929,915,854,943,027,200. No relation is manufactured by the count: field antisymmetry, Bianchi, EOM, integration by parts, trace, and Cayley–Hamilton remain explicit quotient rows. Thus this theorem completes the genuine raw-carrier census, but not the noncommutative physical operator basis or its dual chart.

Our first syntactic quotient removes derivative labels exactly. FORMAL REPAIR. The active derivative slots in the raw carrier are labeled, whereas an operator depends only on the ordered derivative word. Interpreting `derivativeOrder` explicitly as the map from ordered positions to active labels separates every raw derivative presentation into a label permutation and the owner–axis datum at each ordered position. Equality of the latter data is proved equivalent to a unique relabeling of the active derivative slots. Consequently the actual quotient of the original exact dimension-16 subtype is linearly represented by ordered derivative words; no canonical representative is assumed.

This alpha quotient commutes with the literal $H(4)$ action and preserves the product of all active tensor signs. Before any physical identity is imposed, the layout multiplicity falls from 5,361,612,271,200 to 5,030,751, and the corresponding carrier has exactly 21,606,911,019,319,296 elements (`HypercubicDimension16AlphaReduction`). Field-label alpha equivalence and the standard field antisymmetry, Bianchi, EOM, integration by parts, covariant-derivative commutator, trace, and SU(2) Cayley–Hamilton identities remain to be imposed. The commutator identity is listed separately because $[D_\mu, D_\nu]X = [F_{\mu\nu}, X]$ is necessary to identify noncommuting derivative words and is not supplied by Bianchi or integration by parts.

Our initial relation rows are equivariant on the same free module. FORMAL REPAIR. The signed hypercubic action is extended linearly from the alpha-reduced basis, with the complete active tensor sign attached to each basis vector. Three sparse row families are constructed on this module: uniform field-label renaming, swaps of adjacent ordered derivative positions owned by different field factors, and $F_{\mu\nu} + F_{\nu\mu} = 0$. Each family is proved to intertwine exactly with the signed $H(4)$ action (`HypercubicDimension16RelationEquivariance`). The derivative-shuffle site carries explicit adjacency and different-owner conditions. The adjacency condition prevents a nonlocal transposition from reversing hidden same-owner derivatives. Positions owned by the same field are never identified by a shuffle, since their exchange requires the curvature term in the

covariant-derivative commutator. These are equivariance and sparsity theorems; their joint rank has not yet been inferred from the row labels.

Our Bianchi and on-shell EOM rows respect derivative order. Formal repair. Ordered derivative positions are read from outermost to innermost. A derivative is eligible for Bianchi or EOM reduction only when it is innermost among the positions owned by its field factor. The predicate is preserved by $H(4)$. On such a context the three-term row

$$D_a F_{bc} + D_b F_{ca} + D_c F_{ab}$$

and the contracted on-shell row

$$\sum_{\mu=1}^{4} D_\mu F_{\mu\nu}$$

are constructed explicitly and proved equivariant for the same signed action (`HypercubicDimension16DifferentialRelations`). The EOM rows belong only to the on-shell policy. The innermost condition is essential: moving either identity through deeper derivatives without curvature commutators would change the represented operator. No rank or completeness claim is deduced merely from these formulas.

Our integration-by-parts row is an integrated action identity. Formal repair. A globally outermost derivative position is chosen and its owner is varied over every field factor. The sum of those basis rows is the product-rule expansion of an integrated total covariant derivative and is proved equivariant under the signed $H(4)$ action. The outermost condition is preserved by the action and is compatible with the different-owner shuffle rows. This is an action-sector integration-by-parts relation; it is not asserted as pointwise vanishing of a local derivative.

The same module supplies the elementary traceless-letter row: a fixed point of the trace permutation represents a one-letter trace and is killed because every covariantly differentiated field strength is $\mathfrak{su}(2)$-valued and traceless. Disjoint cycles already encode cyclic words without a chosen starting point or ordering of trace factors, while uniform field-label alpha rows identify conjugate labeled presentations (`HypercubicDimension16IBPTraceRelations`). Higher SU(2) trace identities remain separate.

Our covariant-derivative commutator is an exact cross-sector row. Formal repair. A site consists of two consecutive positions in the ordered derivative word of a single field letter; derivatives belonging to other field letters may occur between them in the global word. The construction removes precisely those two positions while retaining the relative order of their complement, adds a curvature letter with the two derivative axes, and inserts it immediately before or after the selected field in the same trace cycle. The resulting four-term row is

$$D_a D_b X - D_b D_a X - F_{ab} X + X F_{ab} = 0.$$

Both curvature insertions have the same canonical dimension as the two derivative terms. The trace-cycle insertion equations, dimension identity, complement-order embedding, active tensor sign, and signed $H(4)$ intertwining law are proved exactly (`HypercubicDimension16CovariantCommutator`). Thus no same-field derivative swap is imposed without its curvature commutator. This theorem constructs the relation operator; it does not infer its rank or the rank of the combined quotient. The higher SU(2) trace identities are supplied next as independent row schemas.

Correction (September 2026): the commutator row needs the Leibniz rule. The four-term row above is a consequence of the field identities only when the selected field letter has no derivatives outside the pair. If $X = D_{w_1} D_a D_b Z$ with a nonempty outer word w_1, the identity is $D_{w_1}(D_a D_b - D_b D_a)Z = D_{w_1}(F_{ab}Z - ZF_{ab})$, and the adjoint Leibniz rule distributes w_1 over both factors,

$$D_{w_1}(F_{ab}Z) = \sum_{S \subseteq w_1} (D_S F_{ab})(D_{w_1 \setminus S} Z),$$

each factor keeping the order of its derivatives. The four-term row keeps only $S = \varnothing$. On an explicit constant SU(2) background, with covariant derivative $\operatorname{ad} A$ and curvature $[A, A]$, the four-field row used below for the field-five separator, whose field carries six outer derivatives, has value 64; it is therefore not a field identity (`HypercubicDimension16CommutatorRowEvaluation`). The corrected row subtracts, for every subset $S \subseteq w_1$, the two insertions of F_{ab} before and after the field in which the derivatives of S have moved onto F_{ab}. It is signed-$H(4)$ equivariant and equals the four-term row when w_1 is empty (`HypercubicDimension16LeibnizCommutator`). Every corrected row vanishes on every gauge background, for every site: the proof identifies each letter as covariant derivatives along an ordered list of positions, and combines $[D_a, D_b]Z = [F_{ab}, Z]$ with the adjoint Leibniz rule over the subsets of w_1 (`HypercubicDimension16LeibnizCommutatorSoundness`). The other families are sound as well: field antisymmetry, the innermost Bianchi row, the traceless one-letter trace and the two SU(2) trace identities vanish on every background, the integrated total-derivative row vanishes after integration, and the equation-of-motion row vanishes on backgrounds satisfying the Yang–Mills equations. Every integrated evaluation therefore vanishes on the corrected relation submodule, off shell on every background and on shell on every on-shell background. The earlier four-field row is a row of the earlier operator, yet lies outside the off-shell corrected relation submodule (`HypercubicDimension16RelationRowSoundness`). The evaluation descends: the value of a relation carrier is its orientation coefficient times the value of its normalized labeled sector, and that value is invariant under field relabeling. Every integrated evaluation is therefore a functional on the field-label orbit space that vanishes on the corrected orbit and joint relation submodules, and it induces a functional on the corrected joint scalar quotient. The orbit column of the earlier four-field row has value 64 on the constant background, so it lies in the earlier orbit relation range but outside the corrected one (`HypercubicDimension16RelationEvaluationDescent`). The same holds at the invariant level, which refutes the earlier joint submodule as a space of relations. Evaluation intertwines the signed $H(4)$ action with rotation of the background, so the Reynolds average of a vector is evaluated by averaging over the 384 rotated backgrounds. The Reynolds-averaged four-field row, a vector of the earlier joint invariant relation submodule, has averaged value $32/3$ on the constant background: the row takes the value 64 on exactly 64 rotated backgrounds and 0 on the others. It is therefore not a joint relation of the corrected operator. The earlier joint submodule, to which the bound 11,558 of the earlier modules refers, contains an invariant vector that is not a consequence of the field identities (`HypercubicDimension16RelationEvaluationEquivariance`). The corrected operator and joint submodule are defined on the same generators (`HypercubicDimension16LeibnizRelationOperator`). With seven or eight source fields a carrier has at most two derivatives, so a commutator site uses both and its field has no outer derivative; the two operators agree on every column of those source bands, and the source-seven and source-eight results below apply to the corrected operator unchanged. The bounds 11,557 and 11,558 below hold for the corrected submodule: the 11,557 rows are unchanged, and every Leibniz term of the four-field separator row misses the separator's orbit, so its coordinate is still $-1/24$ (`HypercubicDimension16LeibnizJointRank`). The finite cochain bridge carries over: the corrected operator followed by the Reynolds average is signed-$H(4)$ equivariant, compatible with the two policies, and maps the generator space onto the corrected joint submodule (`HypercubicDimens`

ion16LeibnizFDIBPCochainBridge). So does the source-band reduction of the profile correction summarized in the abstract. A corrected commutator row with six source fields has one pair of seven-field insertions for each subset S of the outer word; the two insertions of a pair have the same letters and conjugate trace permutations, so each pair cancels as the single pair of the four-term row does. With at most five source fields every term of a corrected row has at most six fields and is invisible to the correction. The full profile correction therefore annihilates every column of the corrected operator outside the source-seven band (HypercubicDimension16LeibnizSourceSix).

A literal seven-field, two-derivative instance proves that this cross-sector coupling survives both exact contextual normalization and simultaneous field relabeling. Its two eight-field curvature insertions are identified by independent fixed-width witnesses with two distinct entries of the checked positive orbit chart. The eight-field part of the corresponding physical-relation column is exactly

$$-e_{10513} + e_{10515},$$

and its signed Reynolds average is nonzero by off-diagonal orbit separation and the nonzero diagonal coefficient theorem. Hence the descended physical relation operator is not block diagonal in field count: the derivative-free eight-field trace quotient cannot replace the joint dimension-sixteen physical quotient without the incoming commutator block (HypercubicDimension16CovariantCompressionWall).

Our joint invariant quotient is fixed before rank computation. Formal repair. Let V be the genuine simultaneous-field-relabel orbit module, $I = V^{H(4)}$ its signed hypercubic fixed submodule, and $R_p \subset V$ the range of the joint physical-relation operator under policy p. We define the invariant relation submodule canonically as

$$J_p = \iota^{-1}(R_p) \subset I,$$

where $\iota : I \hookrightarrow V$ is the inclusion, and define the scalar physical quotient to be I/J_p. Membership in J_p is therefore exactly membership of the underlying invariant vector in the full relation range; the structural definition materializes neither a Reynolds matrix nor a dense relation matrix. The off-shell generator space embeds in the on-shell one without adding an off-shell EOM row, so $J_{\text{off}} \subseteq J_{\text{on}}$ and the on-shell quotient dimension is no larger. This fixes the honest coordinate-free target for sparse certificates; it does not infer either quotient dimension or conditioning (HypercubicDimension16JointPhysicalQuotient).

Our SU(2) **trace rows include genuine multitrace rewiring.** Formal repair. For traceless 2×2 matrices the polarized Cayley–Hamilton identity

$$AB + BA = \mathrm{Tr}(AB)I$$

is proved over an arbitrary commutative ring, together with its trace-context form

$$\mathrm{Tr}(UABV) + \mathrm{Tr}(UBAV) = \mathrm{Tr}(AB)\mathrm{Tr}(UV).$$

On the finite carrier, the right-hand side is constructed by splitting the adjacent A, B letters into a two-cycle and reconnecting the remaining trace word. The three changed permutation edges are proved explicitly.

Adjacent splitting alone does not visibly generate identities which join several trace cycles. Accordingly, for every injection of three cut labels we also construct the standard six-term fundamental 2×2 trace row

$$\mathrm{Tr}A\mathrm{Tr}B\mathrm{Tr}C - \mathrm{Tr}(AB)\mathrm{Tr}C - \mathrm{Tr}(AC)\mathrm{Tr}B - \mathrm{Tr}(BC)\mathrm{Tr}A + \mathrm{Tr}(ABC) + \mathrm{Tr}(ACB) = 0.$$

The identity is proved by matrix evaluation, while its carrier realization is the signed sum over the identity, three transpositions, and two three-cycles acting on the selected outgoing trace edges. Thus its terms honestly join and split cyclic and multitrace layouts. Both row families preserve every axis assignment and active tensor sign and are signed-$H(4)$ equivariant (`HypercubicDimension16SU2TraceRelations`). These are relation and equivariance theorems; no joint rank or quotient conditioning is inferred from the schemas alone.

Our sparse trace-rank replay is kernel checked, and its rational row arithmetic has an exact semantics. Formal repair. A bounded exact enumeration on the genuine eight-field scalar orbit carrier selects 11,556 integer trace-relation rows against the 11,654 orbit coordinates, leaving 98 candidate free columns. The selected rows contain 25,379 nonzero integer entries. Exact elimination produces 27,568 nonzero rational entries, uses only 675 reductions by previous pivots in total, and uses at most 25 such reductions for any one row. The rational replay is split into forty-five 256-row certificates and one 36-row certificate. Lean's kernel checks every reduction and every normalized row using explicit numerator–denominator arithmetic; no dense matrix is materialized and no evaluator axiom is used.

An independent semantic layer now interprets every valid integer pair (n, d) as $n/d \in \mathbb{Q}$. It proves that exact zero tests and cross-multiplication equality are sound, that sparse insertion and cancellation denote vector addition, that the complete fold used for a scaled row subtraction denotes the corresponding subtraction in the rational coordinate module, and that pivot normalization denotes inverse scalar multiplication. Validity is preserved by every one of these operations under the explicit nonzero-denominator hypotheses (`HypercubicDimension16TraceRankSemantics`).

The forty-six checked ranges are now assembled into one typed witness for every selected row. The normalized rows have pairwise distinct maximal pivots and are therefore linearly independent. Strong induction through the exact replay proves that every normalized row lies in the span of the selected raw rows. Since both indexed families have 11,556 members, the selected raw rows themselves are linearly independent over $\mathbb{Q}$, and their rational span has finrank exactly 11,556 (`HypercubicDimension16TraceRankTheorem`).

We also reran the sparse elimination independently modulo the two primes 1,000,000,007 and 1,000,000,009. Each bounded run examined 5,455,784 generated candidate rows and produced 11,556 pivots with 27,568 nonzero normalized-basis entries. The prime-specific coefficients and reduction traces are stored separately in canonical increasing-support order. Lean proves that both moduli are prime and distinct, checks the stored basis and trace shapes, and replays every one of the 11,556 selected integer rows in forty-six chunks at each prime (`HypercubicDimension16TraceRankModularCertificate`). These two prime-field replays are independent arithmetic audits. The exact $\mathbb{Q}$ theorem above remains the semantic rank result, and the modular checks do not by themselves certify exhaustive generation of the physical relation operator.

Correction (September 2026). The August 2026 repacking of these certificate payloads into bounded packed numerals left three defects that no build caught, because the affected modules no longer built on the previous toolchain. The prime-field payloads are packed with 32-bit fields but were decoded with the codec's 16-bit fields, so all 92 modular range checks were false as written; the kernel refutes the first range at each prime, and no selected row replays at width 16 while all 11,556 replay at width 32. Plain `decide` could not unfold the new irreducible payload accessors, so neither the rational chunks nor the modular checks were being checked. On Lean 4.34 the kernel's decoding of the hexadecimal string payloads is quadratic in their length. The port to Lean 4.34.0-rc2 decodes the modular payloads with 32-bit fields, leaving the payloads, the shared codec, and every statement unchanged; it checks the replays with `decide +kernel`; and it routes the string payloads through

explicit tables proved equal to their decodings (`HypercubicDimension16HexPayloadBridge`). Every replay described in this section is now checked by the kernel.

The independently defined target does not use the encoded coordinate table. On the genuine eight-field field-relabel orbit module, we take the span of the literal signed-orbit rows $v - hv$. This span is proved to be exactly the kernel of the Reynolds projector: the forward inclusion uses right invariance of the group average, while the reverse inclusion writes $v - Pv$ as the average of the rows $v - hv$. Consequently the signed coinvariant quotient is linearly equivalent to the previously constructed Reynolds-invariant subspace and has finrank exactly 11,654 (`HypercubicDimension16SignedOrbitCoinvariants`). The encoded orbit chart is proved below to coordinatize this independently defined standard quotient; it does not define the quotient by its own table.

We also constructed independent one-way invariants for the encoded chart. For each field label the first records the eight successive oriented-plane colors around the literal trace permutation, together with all eight return-to-start bits, as a base-twelve rooted code. Both the sorted multiset of the eight codes and the standard symmetric polynomial

$$\prod_{f=1}^{8}(1 + c_f)$$

are proved invariant under simultaneous field relabeling before they are descended to the genuine orbit carrier. Reflections do not change the plane transport, so minimization requires only the 24 unsigned coordinate permutations; the resulting minimum is proved invariant under the full literal hypercubic group (`HypercubicDimension16TraceOrbitSignature`). Distinct minima therefore certify distinct genuine carrier orbits, but no converse or completeness property is assumed.

The fixed-width payload contains 11,654 product-signature minima, all distinct. It stores the values in representative order and strictly sorted order together with mutually inverse rank tables. The kernel replay is partitioned into 182 ranges of 64 representatives and one final range of 6; each range recomputes all 24 unsigned images directly from the checked trace permutation and plane colors. The global certificate separately checks the rank inverse and the sorted table in eleven 1024-entry blocks and one 390-entry tail; the same bounded replay checks all 11,653 adjacent strict inequalities before using the one-way orbit theorem (`HypercubicDimension16TraceOrbitProductSignatureCertificate`). Exhaustivity of the chart remains a separate obligation.

For the stabilizer sign, an even coarser standard invariant suffices. The six plane multiplicities are defined on the labeled carrier, proved invariant under field relabeling, and descended to the genuine orbit carrier. Exact enumeration gives 32,940 unsigned plane-histogram stabilizer pairs and no negative tensor sign; every representative also has even incidence at each coordinate axis. The two-form tensor sign factors exactly into its unsigned permutation part and its pure-reflection part. Accordingly, the bounded certificate checks the 16 pure reflections and the 24 unsigned plane-histogram implications separately for each representative; the factorization then supplies all 384 signed elements. This tests only a necessary condition for an actual stabilizer and assumes completeness of neither the plane histogram, the rooted code, nor the product signature (`HypercubicDimension16TraceOrbitPlaneHistogram`, `HypercubicDimension16TraceOrbitHistogramSignCertificate`).

For each decoded target, take the Reynolds average of its standard coordinate vector. Pairwise orbit separation makes the evaluation matrix of these vectors diagonal, and positive stabilizers make every diagonal entry a strictly positive rational number. The 11,654 Reynolds vectors are therefore linearly independent. Their cardinality equals the independently proved finrank 11,654 of the invariant subspace, so they form a basis; transport through the Reynolds equivalence gives a basis of the standard signed coinvariant quotient (`HypercubicDimension16TraceOrbitBasis`).

This proves that the chart coordinatizes the independently defined quotient rather than defining it by enumeration.

Every selected raw row is now proved to have support inside the finite 11,654-coordinate ambient module. Restriction to that module preserves linear independence, so rank–nullity gives dimension 98 for the quotient by the selected-row span. The 11,556 normalized rows together with the 98 certified free coordinate vectors form a basis of the ambient module. Their free images form a basis of the selected-row quotient, and the corresponding 98×98 analysis matrix is exactly the identity (`HypercubicDimension16TraceFiniteQuotient`).

Every selected encoded row is also identified with an actual Reynolds image of a literal standard trace row. The certificate decodes the stored source term, field relabeling, hypercubic transport, tensor sign, and integer coefficient of each nonzero coordinate. Thirty ranges of 256 witnesses cover the 7680 one-letter traceless rows; sixty ranges of 64 witnesses and one range of 36 cover the 3876 six-term fundamental rows. A separate finite coefficient replay is proved equal to the corresponding semantic finitely supported vectors, so the checked payload cannot introduce a new relation by its encoding. Consequently the 11,556-dimensional selected span is a subspace of the genuine standard trace-relation span (`HypercubicDimension16PhysicalTraceWitnessBridge`).

Independently of that finite selection, the genuine eight-field orbit module carries the three literal standard SU(2) trace schemas: one-letter tracelessness, the polarized Cayley–Hamilton anticommutator row, and the six-term fundamental 2×2 trace row. Their signed $H(4)$ intertwining law is proved before any rank computation. Each of the 98 rational plane-matrix assignments is kernel-checked to be traceless, and ordinary product-of-traces evaluation annihilates every row in all three families, including after arbitrary hypercubic transport (`HypercubicDimension16FieldEightPhysicalTrace`, `HypercubicDimension16PhysicalTraceEvaluationInvariant`). Their full 98×98 evaluation matrix is checked together with an exact rational inverse. The resulting 98 annihilation functionals are independent on the certified free classes, giving the opposite rank inequality to the selected-row inclusion. Therefore the complete literal standard trace-relation submodule has finrank 11,556, its derivative-free eight-field quotient has finrank exactly 98, and the certified free classes form an honest quotient basis (`HypercubicDimension16PhysicalTraceEvaluationCertificate`, `HypercubicDimension16PhysicalTraceQuotient`).

The genuine 384-element Reynolds normalization makes the public evaluation matrix equal to the stored unsigned matrix divided by 24. Its exact dual is consequently 24 times the stored inverse. These duals are proved biorthogonal to the 98 free classes and reconstruct every quotient vector. The sharp coefficient-sup analysis constant is

$$\frac{14739388451343}{30664304},$$

attained by row 30 (`HypercubicDimension16PhysicalTracePublicConditioning`). If the previously isolated fluctuation, cumulant, and recombination hypotheses are transported unchanged to this evaluation-sup norm, the conditioned ledger is

$$\frac{928581472434609}{1742290}.$$

At block factor two its least linear contraction depth is 32, and its least two-source closing depth is 34 (`HypercubicDimension16TraceEvaluationBootstrap`). This last arithmetic is conditional and concerns only the derivative-free trace sector; it supplies neither the derivative-coupled joint quotient nor an interacting Wilson-RG estimate.

Our quantitative restoration theorem is tree-level and constant-Cartan. Formal repair. The derivative-free Cartan octic restriction contains 1287 monomials and exactly 17 signed

$H(4)$ orbit coordinates. Representative evaluation supplies exact duals and the identity conditioning matrix, hence analysis and synthesis both have coefficient-sup factor one. For the actual SU(2) Cartan link field, the normalized Wilson eighth jet is

$$-2a^{12}\sum_{\mu<\nu}F_{\mu\nu}^{8},$$

so division by 8! leaves only the pure-octic coordinate, equal to $-a^{12}/20160$. If $|F_{\mu\nu}|\le M$ and $|a^2F_{\mu\nu}|\le 1$, the actual six-plaquette density obeys

$$\left|\mathcal{W}_a(F)-\sum_{\mu<\nu}F_{\mu\nu}^2\right|\le\frac{5}{8}a^4M^4,$$

and therefore converges to the $O(4)$-invariant quadratic density along every eventually positive lattice-spacing net tending to zero. The corresponding two-block rotation error is at most $(5/2)a^4M^4$.

The 17 coordinates lift to the postponed-integration-by-parts boundary complex; shared-boundary coboundaries cancel under global recombination, and the degree-zero quotient retains rank 17. The newly derived extraction factor is exactly one. Keeping the fluctuation, cumulant, and recombination bounds as explicit hypotheses gives the conditional recursion budget 5544/5; the two-source bootstrap then closes at depth 15 and fails at every depth at most 14. None of these statements constructs an interacting continuum measure, proves uniform Symanzik estimates for quantum correlators, verifies Osterwalder–Schrader reconstruction, or proves a mass gap.

Scope of the original finite gaps. The $\mathbb{Z}_2$ and $\mathbb{Z}_3$ spectral gaps are honest theorems, but they are abelian, single-link, strong-coupling, and finite. They exemplify the finite OS-reconstruction mechanism; they are not lattice approximations to a continuum SU(N) theory, and the accompanying scaling diagnostics show that the physical gap can close under a naive continuum schedule. They establish the machinery, not the target.

The continuum mass gap is correctly open. The continuum endpoint is gated `openGoal` with no claimed support, and the constructive-QFT prerequisites are all marked unrepresented. The formalization makes no claim, in either direction, about the existence of a continuum Yang–Mills mass gap.

D The stopping-scale step: numerical evidence

This appendix retains the earlier bare-Wilson and hierarchical diagnostics, together with the formal crossing calibration. Numerical measurements are distinguished from Lean theorems. The simulations are not integrations of the full generated interactions in §2.

What the step requires. The global bootstrap (v14, Lemma 4.4 and Appendix F.6) keeps $\beta_j\ge\beta_{\min}$ for every $j\le J$, and Appendix F.6 fixes $\beta_{\min}\approx 150$. By v14's own description, the effective measure at the stopping scale is therefore the Wilson measure at coupling $\beta_J\ge\beta_{\min}$, corrected by extracted couplings of size $O(\beta_J^{-(d-4)/2})$ and by activities that are small in the KP norm. Appendix D represents the same measure as a hard-core polymer gas over a block-decoupled

reference measure, a product over J-blocks (its §D.1), and the tree-decay theorem of its §D.5 (the paper's Theorem 3.5) derives

$$\left|\langle F_1; F_2\rangle_J^T\right| \;\le\; C\,\|F_1\|_\infty \|F_2\|_\infty\, e^{-m_J a_J \,\mathrm{dist}_J(S_1,S_2)}, \qquad m_J a_J \;\ge\; c'\kappa,$$

with C and c' depending only on ε_*, the block geometry and the range of the interactions. The bound is therefore uniform in $\beta_J \ge \beta_{\min}$ and in the volume.

The test. We simulated the Wilson action for SU(2) and U(1) on periodic L^4 lattices with $L = 8, 12, 16$. Each measurement follows one heat-bath sweep (Kennedy–Pendleton for SU(2), von Mises for U(1)) and three overrelaxation sweeps; 500 thermalization sweeps precede 4000 measurements (3000 at $L = 16$), and errors are jackknife errors over 40 bins. The main observable is the energy of each 2^4 block of a $b = 2$ step, the sum of its plaquette energies $1 - \frac{1}{N}\mathrm{Re}\,\mathrm{Tr}\,U_p$, through the correlation coefficient $\rho(R)$ of block energies at block distance R along an axis. For near-Gaussian block energies, bounded functions of the standardized block energies, whose sup norms do not depend on β, have covariances proportional to $\rho(R)$; so ρ is what the tree-decay theorem constrains. For U(1) we also measured the transverse photon, $\sin\theta_{bc}$ at momentum $2\pi/L$ along b, and compare its cosh effective mass with the massless lattice dispersion $2\,\mathrm{arsinh}\,\sin(\pi/L)$. The $\beta \to \infty$ values of ρ are exact free-field values on the same tori. The programs are `ym_stopping_scale_mc.c`, `ym_stopping_scale_free_field.py` and `ym_stopping_scale_analysis.py`, next to this document; the last lists the exact runs.

group	β	L	plaquette energy	$\rho(1)$ (free 0.0336)	photon m_{eff} (massless)
SU(2)	150	8	0.005006	0.0344(5)	
SU(2)	150	12	0.005007	0.0335(2)	
SU(2)	150	16	0.005007	0.0334(1)	
SU(2)	2.5	12	0.347985	0.0514(2)	
SU(2)	2.5	16	0.348024	0.0512(2)	
U(1)	150	8	0.001667	0.0335(5)	0.740(5) (0.748)
U(1)	150	12	0.001668	0.0336(2)	0.523(4) (0.512)
U(1)	150	16	0.001668	0.0335(2)	0.386(3) (0.388)
U(1)	2.0	8	0.135167	0.0389(5)	0.749(5) (0.748)
U(1)	2.0	12	0.135174	0.0386(2)	0.509(2) (0.512)
U(1)	2.0	16	0.135181	0.0385(2)	0.387(2) (0.388)
U(1)	0.8	8	0.598049	0.0246(4)	3.59(4)
U(1)	0.8	12	0.597976	0.0239(2)	3.61(3)
U(1)	0.8	16	0.598015	0.0237(1)	3.60(3)

The photon masses are effective masses between times 1 and 2, and between times 0 and 1 at $\beta = 0.8$, where the signal ends.

Findings.

1. At $\beta = 150$, the measured SU(2) plaquette energy and nearest-block correlation agree with the free-field predictions within the reported statistical errors. The plaquette energy agrees with the weak-coupling value $3/(4\beta) + O(\beta^{-2})$, and $\rho(1)$ agrees with the exact free value on all three lattices. The free field's correlations decay as a power law: on the 64^4 torus $\rho(R) \approx 1.9 \times 10^{-3} R^{-8}$, with $\rho(8)/\rho(9) = 2.57 = (9/8)^8$, so the local decay rate per block,

$8\log(1+1/R)$, tends to 0. A power law eventually exceeds any exponential, whatever C and c' are. Within the reach of the simulation ($R \le 4$) the two cannot be told apart: the infinite-volume free-field tail is a separate analytic statement, while extending it to $\beta = 150$ is a perturbative extrapolation.

2. At $\beta = 2.5$, in the confining regime, the short-distance correlations are larger, not smaller: $\rho(1) = 0.051$, and at block distance 2 the 16^4 lattice gives $\rho(2) = 7.7(1.4)\times 10^{-4}$, twenty times the free value. A mass gap governs only the asymptotic decay, so short-distance magnitudes do not separate the two regimes; the tail and the photon do.

3. The weakly coupled U(1) photon measurements are compatible with the massless dispersion: the effective mass follows $2\,\mathrm{arsinh}\,\sin(\pi/L)$ at $\beta = 150$ and $\beta = 2$, while in the confined phase ($\beta = 0.8$) it stays at 3.60 for every L.

4. Two-loop asymptotic scaling places the SU(2) confinement scale ($\beta \approx 2.5$) about 570 steps of $b = 2$ below $\beta = 150$, a length factor of $10^{171.7}$. With the standard lattice values $m_{0^{++}}/\sqrt{\sigma} \approx 3.8$ and $a\sqrt{\sigma} \approx 0.18$ at $\beta = 2.5$, the glueball mass at $\beta = 150$ is about 10^{-172} in lattice units, so a uniform bound $m_J a_J \ge c'\kappa$ at $\beta_J \ge 150$ would conflict with that asymptotic-scaling estimate. This is an extrapolation, not a measurement or a rigorous exclusion at finite coupling.

A structural observation: the abelian test. We found no step of v14 between the bootstrap and the tree-decay theorem that uses a property distinguishing SU(N) from U(1). The one group-dependent input is the drift coefficient, which vanishes for U(1); zero drift only makes $\beta_j \ge \beta_{\min}$ easier to maintain. Read with $G = \mathrm{U}(1)$, the argument would give exponential clustering with a volume-independent mass for the weakly coupled four-dimensional U(1) lattice gauge theory, and so would the area law that v14's Appendix N lists as accessible by the same methods. Both are false: at weak coupling that theory has a perimeter law and a massless photon (Guth 1980; Fröhlich and Spencer 1982). A proof of the crossing step must therefore use a property that separates the relevant non-Abelian trajectory from this Abelian test. The argument does not prove failure for SU(N) at fixed finite coupling. It identifies the passage from a weakly coupled measure to a KP-small gas over a block-product reference as the step requiring additional justification.

A barrier suite, and the hierarchical lattice. A replacement for the stopping-scale step has to fail wherever its conclusion is false: for weakly coupled U(1) in four dimensions, which has a massless photon (Guth 1980; Fröhlich and Spencer 1982), and for weakly coupled SU(2) in five dimensions, which lies beyond a bulk phase transition (Creutz 1979). It also has to be consistent with asymptotic freedom, which places the four-dimensional mass at $\beta = 150$ near 10^{-172} (finding 4). The diamond hierarchical lattice, on which the Migdal–Kadanoff recursion with block factor 2 is exact, is a natural testbed, because the whole flow can be computed there. One step raises the plaquette weight to the power 2^{d-2} and then raises its normalized character coefficients to the fourth power. We iterated it on character coefficients in 110-digit arithmetic (`ym_hierarchical_flow.py`). In five dimensions both groups flow to weak coupling: β doubles at each step for U(1), and $1/t$ roughly doubles for SU(2), with t the heat-kernel time. In four dimensions SU(2) crosses to strong coupling: at weak coupling $1/t$ decreases by 1/8 per step, so $\beta \approx 2/t$ decreases by 0.25, against $4N^2 b_0 \log 2 = 0.2575$ on the cubic lattice. But U(1) crosses as well. Its coupling decreases at every β, by 1.1×10^{-5} per step at $\beta = 1$, 6.1×10^{-12} at $\beta = 2$ and 1.0×10^{-31} at $\beta = 5$; the drift is $-C\beta^2 e^{-a\beta}$ with $a = 15.7914$, which agrees with $8\pi^2/5$ to all digits computed. These calculations suggest confinement on the four-dimensional U(1) hierarchical

model at every finite coupling, with a doubly exponential weak-coupling correlation length. The all-coupling assertion is an extrapolation of the computed recursion, not a Lean theorem or a uniform numerical certificate. The hierarchical lattice fails the abelian test in four dimensions: a weak-to-strong flow proved there cannot be the mechanism on the cubic lattice unless the proof uses an ingredient that the hierarchical lattice lacks.

The rigorous calibration concerns the free lattice Maxwell field, whose field strength is gauge invariant. Its identification with an appropriate $\beta \to \infty$ scaling limit of compact U(1) is a separate limit theorem, not supplied by this formalization. The Fourier symbol of its F_{01} two-point function, $(\hat{k}_0^2 + \hat{k}_1^2)/|\hat{k}|^2$ with $\hat{k}_\mu^2 = |e^{2\pi i k_\mu} - 1|^2$, equals 1 on the punctured k_0-axis and 0 on the punctured k_2-axis, and it is continuous away from the origin. It therefore agrees almost everywhere with no continuous function, whereas an absolutely summable Fourier series converges uniformly to a continuous one. So the two-point function is not absolutely summable and admits no bound $Ke^{-\mu|n|_1}$ with $\mu > 0$ (`FreeMaxwellNonClustering`). The same holds in every dimension $d \geq 3$, in particular in five, and for the square of the two-point function; for the free SU(2) field, with three independent colour components, the connected two-point function of the gauge-invariant observable $\sum_a (F_{01}^a)^2$ is 6 times this square by Wick's theorem. No family of two-point functions converging pointwise to the free one satisfies an exponential bound with uniform constants. The lemma behind it applies to every two-point function whose symbol has direction-dependent limits at zero momentum. It does not cover finite β, where the masslessness of the photon is the theorem of Fröhlich and Spencer.

The crossing input, stated and calibrated. The missing input can be stated precisely and tested. Call a family of block lattices, indexed by the volume, *crossing* if at one block scale every effective measure is a hard-core polymer gas over a block-product reference measure whose activity bounds satisfy $\sum_{\gamma \ni x} k_\gamma e^{4|\gamma| + \kappa \operatorname{diam} \gamma} \leq \varepsilon$ at every block x, with $\varepsilon e \leq 1$, $k_\varnothing = 0$ and $\kappa > 0$, all independent of the volume. This is the regime v14's tree-decay theorem presupposes, with the Kotecký–Preiss norm written out. The Lean record `CrossingInput` contains the bounds on the activities; that an actual stopped Wilson or RG measure admits such a representation is part of what the route has to supply. Under the bounds, for bounded measurable observables F_1, F_2 on block sets S_1, S_2 with S_1 nonempty,

$$\big|\langle F_1 F_2\rangle - \langle F_1\rangle\langle F_2\rangle\big| \;\leq\; 3\,\|F_1\|_\infty \|F_2\|_\infty\, e^{3(|S_1|+|S_2|)}\, e^{-\kappa\, d(S_1,S_2)},$$

uniformly in the volume and the polymer set (`PolymerGas.LatticeClustering`, `CrossingInput`). The proof uses no cluster expansion: Dobrushin's criterion, implied by Kotecký–Preiss, gives non-vanishing partition functions and two-sided single-removal bounds by induction on the polymer set; the effect of removing a far set of polymers is bounded by an exponential supersolution of an influence recursion; and the truncated two-point function decomposes, by conditioning on the support of F_1, into families that stay away from S_2 and families that bridge to it (`PolymerGas.Basic`, `PolymerGas.Decay`, `PolymerGas.Measure`, `PolymerGas.Clustering`, `PolymerGas.Lattice`). The input is calibrated by the barrier. Take bounded observables on block sets of bounded size whose separation, for each fixed displacement k, eventually exceeds $\alpha|k|_1 - \beta$ with $\alpha > 0$ as the volume grows. No crossing family can have two-point functions of such observables whose limits dominate the field-strength two-point function of the free lattice Maxwell field, since the uniform bound would pass to the limit (`CrossingInput.false_of_tendsto_dominating`). On the tori the geometric hypotheses hold for single links, with slope $1/d$ (`CalibrationSeparation`), so only the convergence and the domination of the two-point functions remain as hypotheses. For $0 < t \leq 1/2$ the expression $e^{-t\sigma^2}(e^{tC(n)} - 1)$, the truncated two-point function of the phase $e^{i\sqrt{t}F_{01}}$ and its conjugate for a

centred Gaussian field of covariance C, controls $|C(n)|$ up to the factor $2e^{t\sigma^2}/t$, because $|C(n)| \le 1$. For the free field itself the calibration is complete. Let the centred Gaussian field F_{01} on the tori have the periodic Maxwell covariance; its covariances converge to the free ones as the volume grows (a Riemann-sum argument for the bounded symbol, continuous away from zero). Then no crossing family reproduces its two phase marginals $e^{\pm iF_{01}/2}$ and their joint expectation at every volume (`FreeFieldNoCrossing`). Not formalized are the $\beta \to \infty$ limit that would put weakly coupled four-dimensional U(1) and the abelian components of five-dimensional SU(2) near this field, and the persistence of weak coupling at every scale for those theories. For four-dimensional SU(2), reaching a regime with these bounds by running the coupling from weak to strong is the proposed mechanism; establishing it is the open crossing.

The input is not vacuous. Take the tori $(\mathbb{Z}/L)^d$ with $L \ge 3$, independent link variables, and any measurable plaquette potential ϕ with $0 \le \phi \le 2$; the Wilson action $1 - \operatorname{Re}\operatorname{tr} U_p/N$ with Haar measure is one. The connected plaquette expansion writes $\prod_p e^{-\beta\phi_p}$ exactly as a polymer density over link sets. For $0 \le \beta \le \beta_0(d) = e^{-18}/(4(4d^2+1)(16d^2+1)^2)$ its activities satisfy the per-block smallness with rate 1 and $\varepsilon = e^{-1}$ at every volume, and the Wilson expectations cluster exponentially, uniformly in the volume (`StrongCouplingCrossing`). The ingredients are the expansion (`PolymerGas.PlaquetteExpansion`), the bound $(\Delta+1)^{2n}$ on connected sets of n plaquettes through a given one in a graph of degree at most Δ (`PolymerGas.ConnectedCount`), and the explicit threshold (`PolymerGas.PlaquetteSmallness`). The threshold is far from sharp. This theorem supplies a nonvacuous strong-coupling endpoint for the route; the weak-to-strong entrance and continuum consequences require the additional inputs in §6.

Candidate mechanisms for the crossing. What four-dimensional SU(N) has and four-dimensional U(1) and five-dimensional SU(2) lack is a short list: a positive one-loop coefficient in the dimension where the coupling is dimensionless, gluon self-interaction, a finite centre with $\pi_1(\mathrm{SU}(N)/\mathbb{Z}_N) = \mathbb{Z}_N$, and instantons, $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$. Only the first is specific to four dimensions; the others are present in five as well. We screened candidate mechanisms built on them against the barrier suite, the calibration at $\beta = \infty$, and two further theorems, one of which can be tested at finite coupling. In the order tested:

1. *Thresholds on one-scale marginals*, such as the mean block energy, the probability that a block exceeds the small-field cutoff of a multiscale analysis, or the law of a blocked plaquette. Refuted as sufficient conditions, formally, whenever the admitted one-site law has a bounded observable of nonzero variance: the product measure and the diagonal measure, in which every site carries the same μ-distributed value, have the same marginal at every site, but the first has no correlations at distinct sites and the second has the covariance at every pair, whatever their distance (`MarginalBlindness`). The stopping-scale test of this section is of this kind: it can refute a crossing but not certify one.

2. *Reflection positivity, infrared bounds, chessboard estimates.* Refuted: the free lattice Maxwell field and weakly coupled four-dimensional U(1) are reflection positive and do not cluster. These methods prove long-range order, not decay (Fröhlich, Simon and Spencer 1976; Fröhlich and Spencer 1982).

3. *Decimation bounds and their interpolation.* Refuted by the abelian test on the hierarchical lattice. There the twist free energy of one coarse plaquette, $F = -\log f(-1)/f(1)$, with f the effective plaquette weight and -1 the centre element (the flux π for U(1)), behaves as follows (`ym_hierarchical_flow.py`). In four dimensions, for SU(2) from $t = 0.5$, it falls by about 1.2 per step, $15.4 \to 14.2 \to \cdots \to 1.61$, and then collapses: 0.665, 0.063, 7.9×10^{-6},

2×10^{-21}. For U(1) from $\beta = 1.5$ it stays at 6.709 to four digits over twelve steps, but its coupling falls at the exponentially small rate found above, so it collapses as well. In five dimensions it doubles at every step for both groups. An argument that derives the collapse without using the size of the drift proves it for U(1) too, which is false on the cubic lattice.

4. *Centre decompositions.* Writing SU(2) link variables as SO(3) variables together with $\mathbb{Z}_2$ plaquette variables uses only a central $\mathbb{Z}_2$, which U(1) also has, with $\mathrm{U}(1)/\mathbb{Z}_2 \cong \mathrm{U}(1)$. Refuted as a mechanism by the abelian test: what separates the groups is the dynamics, not the decomposition.

5. *The twist free energy.* Survives the barrier suite numerically, and is tied to the crossing input by a theorem. On the torus $(\mathbb{Z}/L)^d$ let the plaquettes $(x; 0, 1)$ with $x_0 = x_1 = 0$, one in every $(0, 1)$-plane, carry the coupling b and all others β. Then $Z(-\beta)$ is the partition function with a twist through every $(0, 1)$-plane, and

$$F(L) = -\log \frac{Z(-\beta)}{Z(\beta)} = \int_{-\beta}^{\beta} \langle E_{\mathrm{stack}} \rangle_b \, db,$$

with E_{stack} the sum of $\mathrm{Re}\,\mathrm{tr}\, U_p/N$ over those plaquettes. In a polymer representation built from plaquette factors, a change of variables that multiplies link variables by the centre element moves the twist, and polymers away from the twist do not see it, so only the integrated activities of polymers that meet every position of the twist change (`TwistDiagnostic.differ_of_movable`). If two representations under common Kotecký–Preiss bounds differ only on polymers that meet a cut Y and have diameter at least D, then $\left|\log \frac{|Z_{\mathrm{tw}}|}{|Z|}\right| \leq 2|Y|\varepsilon e^{-\kappa D}$ (`PolymerGas.Twist`). So a twist-compatible crossing family, with cuts of polynomial size and separations growing linearly in L, has $F(L) \to 0$, and a family whose twist free energy stays away from zero admits no such twist-compatible representation (`TwistDiagnostic`). That moving the twist is such a change of variables, and that distant polymers do not see it, are hypotheses there. For the connected plaquette expansion of Wilson-type measures with a central involution z (the centre element -1 of SU(2)), a z-invariant probability measure on link variables and a measurable plaquette function with values in $[0, 2]$ they are proved, and at strong coupling the twist free energy of the Wilson measure, with both partition functions genuine Wilson integrals, tends to zero as the volume grows (`StrongCouplingTwist`). The calibration at $\beta = \infty$ needed a limit; this one can be measured at the coupling of interest. Taking direction 0 as time, the $\mathbb{Z}_2$ Fourier transform in the twist gives the free energy $-\log\tanh(F/2)$ of one unit of electric flux along direction 1 ('t Hooft 1979), so $F \to 0$ from above means that this flux costs unboundedly much. This relation and the thermodynamic-integration formula above are standard identities used for the numerics, with $Z(b)$ as defined there; they are not part of the formal statements.

In one respect the twist is specifically non-abelian. In U(1), for $\beta \geq 0$, every twisted configuration has action at least $2\beta L^{d-4}$ in the twisted planes, and the uniform one at most $\frac{\pi^2}{2}\beta L^{d-4}$ (`TwistCost`). In SU(2) the matrices $i\sigma_1$ and $i\sigma_2$ anticommute, so with $i\sigma_1$ on the links of direction 0 in one column, $i\sigma_2$ on those of direction 1 in one row and the identity elsewhere, every plaquette of the twisted planes has maximal weight and their twisted action vanishes (`TwistEater`); extended to the other directions, with the identity on the remaining links, it makes every other plaquette the identity as well. So the classical minima give SU(2) no action cost of the U(1) kind: whatever twist free energy there is comes from the integrals themselves, including their fluctuations. We measured $F(L)$ by heat-bath Monte Carlo, integrating in b

by Gauss–Legendre quadrature (`ym_vortex_free_energy.c`, `ym_vortex_driver.py`); errors are one standard deviation:

	β	$L=4$	$L=6$	$L=8$	$L=10$
SU(2), $d=4$	2.3	0.881(26)	0.106(39)	0.023(48)	
	2.5	1.717(27)	1.197(39)	0.635(53)	0.384(89)
	3.0	2.848(29)	2.559(45)	2.409(74)	
U(1), $d=4$	0.9	0.016(24)	0.017(24)	0.012(33)	
	1.1	2.810(14)	2.831(22)	2.859(36)	2.848(46)
	1.5	5.144(16)	5.187(28)	5.290(41)	
SU(2), $d=5$	1.5	−0.001(35)	−0.014(68)		
	1.8	5.173(54)	5.98(14)	6.14(16)	
	2.0	5.962(66)	6.74(12)		

In the confined phases the integrand is smooth, and 16-point Gauss–Legendre quadrature suffices. In the ordered phases it changes sign across a window near $b = 0$ whose width shrinks like L^{2-d} and whose position carries most of F; there we used a composite rule with 24 to 32 nodes in that window. The plain 16-point rule underestimated F there by up to 20% at $L = 8$.

In the confined phases, four-dimensional U(1) at $\beta = 0.9$ and five-dimensional SU(2) at $\beta = 1.5$, below its bulk transition, F vanishes within errors. In the Coulomb phase of four-dimensional U(1) it does not depend on L within errors: about 2.84 at $\beta = 1.1$ and 5.2 at $\beta = 1.5$, that is $\frac{\pi^2}{2}\beta_R$ with $\beta_R \approx 0.58$ and 1.05. For five-dimensional SU(2) above the transition it stays near six and does not decrease: 5.17, 5.98 and 6.14 at $\beta = 1.8$ for $L = 4$, 6 and 8, and 5.96 and 6.74 at $\beta = 2.0$ for $L = 4$ and 6. For four-dimensional SU(2) it decreases with L at every coupling: at $\beta = 2.3$ from 0.88 to 0.02 between $L = 4$ and $L = 8$, at $\beta = 2.5$ from 1.72 to 0.38 between $L = 4$ and $L = 10$. At $\beta = 3.0$, where these volumes are small in physical units, it is of order one, far below the $\frac{\pi^2}{2}\beta \approx 14.8$ that a classical cost would give. So the diagnostic separates four-dimensional SU(2) from both barrier theories, in which F stays away from zero, as is compatible with failure of the twist-compatible crossing input. The implication is only necessary: failure of crossing alone does not force nonvanishing twist free energy.

6. *The running coupling.* Twist eaters exist in five dimensions too, so they do not by themselves make F vanish. In four dimensions the coupling runs, and the data follow it: F decreases slowly at $\beta = 3.0$, from 2.85 to 2.41 between $L = 4$ and $L = 8$, and falls to zero at $\beta = 2.3$, where the same L are larger in physical units. In five dimensions weak coupling persists and F does not decrease. This is consistent with a flow in which F decreases with L at every coupling and vanishes once L is large compared with the confinement scale; it is evidence, not proof. Vanishing twist free energy would discharge a necessary diagnostic, not the crossing criterion itself. A proof of the required activity and observable bounds on the weak-coupling trajectory remains open.

7. *Assessed, not tested here.* Instantons: a dilute instanton gas produces no area law, and its integral over sizes is governed by the same running coupling. The Gribov horizon: it is specific to non-abelian groups, but it is formulated after gauge fixing, and no route is known from gauge-fixed propagators to clustering of gauge-invariant observables.

The constants. Two of v14's displayed constants were discussed in §A.2: the two ledgers for C_1 and the extraction-projection bound. A third is the drift coefficient. v14 writes $\beta_{j+1} =$

$\beta_j - 4Nb_0 \log b + E_j$ with $b_0 = 11/(3 \cdot 16\pi^2)$ and states that this matches $\beta(g) = -b_0 N g^3$ under $\beta = 2N/g^2$. Those conventions give $d\beta/d\log\mu = 4N^2 b_0$, hence a drift of $4N^2 b_0 \log b$ per step: 0.2575 rather than 0.1288 for SU(2) and $b = 2$, in agreement with the standard SU(2) lattice scale $a\Lambda_L \propto e^{-3\pi^2\beta/11}$. The computation is checked in Lean: along any one-loop flow a block step lowers $2N/g^2$ by exactly $4N^2 b_0 \log b$, and for SU(2) and $b = 2$ this lies between 0.2575 and 0.2576 (`CouplingDrift`). This changes the choice of β_0 but no conclusion.

Scope. The finite-volume data test the bare-Wilson approximation to the proposed stopping description. They do not prove a failure at fixed finite coupling uniformly over volume, nor a failure of the exact generated RG trajectory. The free-field no-go is a separate rigorous calibration. Completing the programme requires control of the actual generated measure and transported observables through the weak-to-strong transition, as specified in §6.

E Detailed finite-certificate record

Machine-checked since the first version of this report

A second formalization round closed several items that the first version listed as open, and sharpened others into theorems:

- **The two-source bootstrap threshold.** Linear contraction ($\Lambda < 1$) is *not* sufficient for the manuscript's bootstrap; the scalar induction closes iff $a = C\, b^{3-d_{\max}} \le 1/3$ (`scalar_bootstrap_two_sources`). At $C = 2224$, $b = 2$: depth 14 fails even linear contraction, depth 15 contracts linearly but the bootstrap *fails* (`not_twoSourceBootstrapSlack_2224_two_fifteen`), depth 16 closes. The true reason for $d_{\max} = 16$ is the 1/3 threshold, not parity of the depth — and the threshold is *dynamical*: an explicit nonlinear toy orbit realizes the recursion with equality and exits the ε^*-ball in one step at depth 15 (`toyRG_ball_escape_two_fifteen`, `benToyRGBootstrapRealization`).

- **The extraction realization and transfer obstruction.** The manuscript's literal displayed sum is ≥ 3 for every $n \ge 1$ and never ≤ 2 (`three_le_literalExtractionOperatorSum`, `not_literalExtractionOperatorSum_le_two`). On an explicit finite Wilson-block polynomial algebra, coefficient-majorant truncation is norm one and gauge invariant (`WilsonBlockMajorant`). A matrix column-sum theorem transfers through explicit basis and dual jets; v14 does not provide these for $P_{\text{ext}} \le 16$, and a skew-projection counterexample proves that no bound follows from the range alone (`ExtendedExtractionTransfer`). The source-certificate audit further proves that F.4 and O.9 select different EOM relation policies and records the absent lattice representatives, quotient basis, dimension-16 jets, biorthogonality, and conditioning rows (`Dimension16WilsonExtractionCertificate`).

- **The support-growth dichotomy.** The two collar-growth statements the manuscript makes are inequivalent, and both branches are now tight theorems: a coarsen-then-thicken recurrence $r_{j+1} \le r_j/b + A$ yields the uniform bound $r_0/b^J + Ab/(b-1)$, while a thicken-only recurrence $r_{j+1} \le r_j + A$ yields exactly $r_0 + AJ$ (`support_growth_dichotomy`).

- **Gap ⇔ clustering, finite class.** Reflection positivity, self-adjointness, and a q-contraction on the vacuum complement give exponential decay of truncated two-point functions at rate $-\log q$ (`abs_truncatedTwoPoint_le_exp`); conversely, RP plus per-vector moment decay $\langle v, T^n v\rangle \le C_v q^n$ on a dense subset gives the gap (`eigenvalue_abs_le_of_moment_decay`)

— note the *per-vector* constants: only the rate must be uniform, which weakens what the manuscript's Assumptions K.7/M.5 need to supply. A regression records that an additive t-independent tail defeats the conclusion (`moment_decay_additive_defeat`), so large-field remainders must decay, not merely be small. On lattice tori with any finite gauge group and a class function ψ even under inversion, with $0 \le \psi \le 2$, as plaquette potential, the transfer matrix at $0 \le \beta \le \beta_0$ (the strong-coupling threshold in the total dimension) has a positive leading eigenvalue λ_0 with $|\lambda_j| \le e^{-m}\lambda_0$ for every other eigenvalue, where $m > 0$ does not depend on the spatial volume (`StrongCouplingTransferGap`). Wilson expectations are exact cyclic products of the transfer matrix; the time-slice clustering that the strong-coupling crossing input gives, with a rate independent of both extents and a prefactor that grows with the spatial volume, bounds the excess of squared traces over traces of squares, and the finite-dimensional spectral theorem turns that bound into the gap.

- **A non-abelian instance.** The quaternion group $Q_8 \subset SU(2)$ single-link model: machine-checked non-commutativity ($ij = k \neq -k = ji$), reflection positivity, spectral gap carried by the four-dimensional spin isotypic component, and clustering; together with the $\mathbb{Z}_2$ instance this discharges the lattice-side inputs (`massGap`, `reflectionPositive`, `clustering`) of the finite continuum-gate conditional (`z2_continuumMassGap_of_reconstructionMachine`).

Machine-checked in the third round

A third round moved from schematic carriers to the exact SU(2), $b = 2$ Wilson block (§C) and decided the extraction stage on it:

- **Hard truncation refuted intrinsically.** The Cayley–Hamilton relation $a^3b - 4a^4 + a^5 = 0$, with monomial dimensions $(16, 16, 20)$ and the explicit -32 evaluation, proves representative-wise canonical truncation is not well-defined on the evaluated trace-word quotient (`SU2CayleyHamiltonTruncationObstruction`).

- **Exact norm-below-two extractors constructed.** The 21-coordinate associated-graded jet chart has extraction constant 131071/65536 and the one-plaquette trace quotient has 31/16, both certified with explicit bases, identity jet matrices, and biorthogonal duals; the conditional ledger value $90832203/40960 \approx 2217.6$ closes the two-source bootstrap at depth 16 and fails at depth 15 (`SU2AssociatedGradedExtraction`, `SU2IndependentTraceBasis`).

- **A certified dimension separation.** The full graded chart has dimension 12,875,774,670 and admits no linear equivalence onto any space of dimension ≤ 1851, so v14's schematic operator count requires an invariant quotient the manuscript does not construct (`SU2AssociatedGradedExtractionDecision`).

- **The (F, D) transfer socket is uninhabited as written.** The dimension-18/radial-degree-6 descendant and the integration-by-parts/block witness $\partial^2 \mathrm{Tr}(F^2)$ jointly refute the faithful transfer sockets for both relation policies and all RG-compatible extensions (`v14_fullFD_quotientTransfer_noGo`).

Machine-checked in the fourth round

A fourth round constructed the derivative-aware boundary-cochain scaffold and then decided whether its dimension-16 coordinate socket could be filled by the manuscript's Lorentz-only operator census:

- **The repaired filtration and boundary bookkeeping are mechanical.** The canonical-dimension filtration, boundary-aware cochain coordinates, coefficient projection, and conditional scalar bootstrap are machine-checked. Their analytic Wilson-RG factors remain explicit inputs.
- **The actual-lattice O.9 census is refuted at dimension 8.** The exact SU(2) Wilson observable, Cartan chart, fourth-jet calculation, and rational rotation witness prove that the hypercubic quartic jet is not Lorentz invariant (`SU2LatticeFDCensusNoGo`).
- **Both relation branches and six same-target variants are closed.** Neither the off-shell O.9.2 policy nor the on-shell F.4.3 policy admits a complete Lorentz-only certificate. EOM, Bianchi/IBP, cyclic/multitrace, SU(2) Cayley–Hamilton, and postponed-IBP variants share the same uninhabited spanning core.
- **The live fork is explicit.** One must either enumerate an $H(4)$-complete finite-spacing basis and rederive its duals and constants, or prove a Symanzik/continuum projection with quantitative decay of the Lorentz-breaking artifact coordinates. No numerical factor transfers automatically from the refuted census.

Machine-checked in the fifth round

The first branch of the live fork was pursued directly:

- **The exact hypercubic group and finite raw syntax are mechanical.** Signed coordinate permutations form a concrete group of order 384. The raw surface records field and derivative indices, trace cycles, derivative ownership/order, canonical dimension through 16, scalar/pseudoscalar characters, and separate relation-family labels. A label still supplies no relation row by itself (`V14HypercubicFDCensus`).
- **The unpadded dimension-8 quartic census is complete.** The 126 Cartan quartic monomials have checked signed-orbit certificates. Four surviving orbit functions form a basis, representative evaluation supplies exact duals, and the conditioning matrix is the identity (`V14HypercubicQuarticCensusCertificate`, `V14HypercubicQuarticBasis`).
- **The actual Wilson and artifact coordinates are certified.** The Wilson fourth jet is exactly coordinate vector $(6, 0, 0, 0)$. The $O(4)$-invariant function subspace is proper, and the pure-fourth polynomial is nonzero in its quotient (`V14HypercubicQuarticWilsonBridge`).
- **The padded dimension-16 extension is refuted.** Two distinct fixed-width encodings of the empty active syntax differ only in inactive padding. Any active-syntax-faithful basis gives them equal rows, which is incompatible with the right-inverse identity required by `ExactCensusCertificate` (`V14HypercubicFDCensusPaddingNoGo`).
- **The corrected carrier is specified without padding.** The obstruction identifies finite unpadded/dependent (F, D) syntax, or a checked padding quotient, as the required carrier on which the $H(4)$ action and relation rows must descend. No dimension-8 constant transfers automatically. The unpadded construction and its raw character census are supplied in the results below; the relation quotient remains separate.

Machine-checked on the unpadded dimension-16 carrier

All constructions in this subsection are ours and are independent of any claim that the manuscript already contains the needed census.

- **The counted carrier is the actual active syntax.** Exact dimension-16 shapes are classified by $f \in \{0, \ldots, 8\}$ with $d = 16 - 2f$. A checked equivalence identifies the corresponding subtype of the unpadded raw (F, D) syntax with the separated layout–rank-16 tensor presentation, preserves the literal action and tensor sign, and induces a linear equivalence on the covariance submodules. Its exact cardinality is 23,027,949,358,636,282,675,200 (`HypercubicUnpaddedFDSyntax`, `HypercubicRawFDDimension16Census`).

- **Derivative-label alpha equivalence is quotiented exactly.** The raw derivative order is separated into a disposable active-label permutation and the owner–axis word indexed by ordered positions. Equality of ordered words is equivalent to a unique relabeling; the quotient of the original exact carrier is explicitly equivalent to the reduced carrier, equivariant under $H(4)$, and tensor-sign preserving. Its exact layout and carrier counts are 5,030,751 and 21,606,911,019,319,296 (`HypercubicDimension16AlphaReduction`).

- **The first sparse relation operators are equivariant.** Uniform field-label renaming, adjacent different-owner derivative shuffles, and field-strength antisymmetry are explicit rational rows on the same free module. Their signed $H(4)$ intertwining laws are proved before any rank claim; same-owner derivative swaps are excluded (`HypercubicDimension16RelationEquivariance`).

- **The standard rows form one equivariant physical-relation operator.** Our off-shell policy combines antisymmetry, innermost Bianchi, globally outermost integrated IBP, traceless one-cycles, the covariant commutator, and the two ordinary SU(2) Cayley–Hamilton trace consequences; our on-shell policy additionally enables innermost EOM. The signed source and target actions are rational $H(4)$ representations, the joint operator intertwines them, and its image is invariant. The same statements hold after the exact antisymmetry/derivative-localization quotient. This is an equivariance and assembly theorem, not yet a physical-rank claim (`HypercubicDimension16PhysicalRelationOperator`).

- **The physical-relation operator reaches the genuine field-label orbit carrier.** Our ordinary simultaneous field-permutation quotient of the oriented local carrier is lifted to a rational linear map. Explicit simultaneous-renaming difference rows span exactly its kernel, and the induced quotient is linearly equivalent to the orbit module. Its signed $H(4)$ action is a representation, the quotient map intertwines the contextual and orbit actions, and composing it with the joint physical-relation operator leaves an invariant image. Thus the operator target is now the same literal finite orbit carrier whose character was counted above; no sectorwise rank or conditioning is inferred here (`HypercubicDimension16PhysicalOrbitOperator`).

- **The descended operator has an exact sparse field-sector band.** Our source-field function is defined on every standard relation generator. Every raw row is supported in that field sector, except that the two curvature-insertion terms of the covariant commutator lie in the successor sector. Explicit rational field filters commute with both the contextual normalization and the simultaneous field-relabel quotient. Consequently each column of the physical operator on the genuine orbit carrier is supported in its source sector and, when present, its immediate successor. This is a block upper-bidiagonal incidence theorem, not a rank or conditioning result (`HypercubicDimension16PhysicalRelationBlocks`).

- **The honest eight-field scalar invariant space has dimension** 11,654**.** Our signed permutation representation is defined directly on genuine simultaneous-field-relabel orbits. Its 384-term Reynolds average is proved to be the projector onto the fixed subspace, and the trace of each action is the literal signed fixed-orbit character. The exact conjugacy-compression identity therefore gives scalar invariant finrank 11,654 in the $(f, d) = (8, 0)$ sector, without choosing orbit representatives or inserting a padded carrier (`HypercubicDimension16PhysicalInvariantSpace`).

- **Bianchi and EOM preserve noncommuting derivative order.** The cyclic Bianchi row and the contracted on-shell EOM row are supplied only at an innermost derivative of the selected field word. Their signed $H(4)$ intertwining laws are checked, and EOM remains disabled in the off-shell policy (`HypercubicDimension16DifferentialRelations`).

- **Integrated IBP and traceless one-cycles are explicit.** The total-derivative product-rule row is supplied at a globally outermost derivative position, and a one-cycle of the trace permutation is killed by tracelessness. Both row families are signed-$H(4)$ equivariant (`HypercubicDimension16IBPTraceRelations`).

- **The covariant-derivative commutator crosses sectors honestly.** Two consecutive derivatives of one field word are removed in complement order, an F_{ab} letter is inserted on either side of that field in its trace cycle, and the four-term commutator row preserves canonical dimension and active tensor sign. Its signed-$H(4)$ intertwining law is checked. For an explicit seven-field, two-derivative column, the two successor terms occupy distinct checked orbit coordinates, its exact eight-field component is nonzero, and its Reynolds projection remains nonzero. Thus the physical operator is provably not block diagonal in field count (`HypercubicDimension16CovariantCommutator`, `HypercubicDimension16CovariantCompressionWall`).

- **The** SU(2) **trace identities rewire the actual trace permutation.** The traceless anticommutator row splits an adjacent pair into a two-cycle and reconnects its context. The six-term fundamental 2×2 trace row additionally joins and splits distinct cycles through three selected edges. Both semantic matrix identities and both signed $H(4)$ intertwining laws are checked (`HypercubicDimension16SU2TraceRelations`).

- **The derivative-free eight-field standard trace quotient is complete.** The independently checked orbit chart has 11,654 coordinates. A literal 11,556-row standard trace subfamily is linearly independent, while 98 ordinary traceless-matrix evaluations annihilate every standard trace row and have an exact rational inverse on the free classes. Hence the complete standard trace submodule has finrank 11,556 and its quotient has finrank 98. The free classes form a basis with explicit public duals. In genuine Reynolds normalization the sharp row-ℓ^1 dual constant is 14739388451343/30664304, attained at row 30. This is the derivative-free trace quotient, not the derivative-coupled joint physical quotient (`HypercubicDimension16PhysicalTraceQuotient`, `HypercubicDimension16PhysicalTracePublicConditioning`).

- **The first honest cycle-index compression target remains enormous.** For our explicitly defined cycle-index expression—oriented two-form letters, one ordered local derivative word per field, and simultaneous unlabeled field/trace-cycle data—the eight field sectors have identity specialization 47,231,161,854. Its scalar and pseudoscalar $H(4)$ Reynolds values are 246,136,792 and 246,121,092. The literal 384-element action variable is reduced to fourteen power-trace classes only after a checked axis-recurrence/group-power bridge, all traces

through exponent 16, determinant parity, class assignments and multiplicities, the cycle-letter recursion, and the final Reynolds-average equality. These numbers impose none of Bianchi, EOM, IBP, the covariant commutator, or the matrix trace identities. A concrete finite labeled carrier now contains exactly a trace permutation and, on each field label, an oriented two-form plus a finite ordered derivative-axis word of the prescribed total degree. Genuine field-relabel and $H(4)$ actions on this carrier commute; the tensor-sign cocycle is field-relabel invariant, descends to the orbit quotient, and retains its group law. The character theorem identifying that orbit carrier with the displayed cycle-index expression is proved by the assembly described below. For the derivative part, stable projection of a global owner–axis word to one ordered axis word per owner is now surjective with a canonical owner-by-owner right inverse; its semantic quotient is exactly the finite local-word family, commutes with field relabeling and $H(4)$, and every already-defined adjacent different-owner shuffle row lies in its kernel. A head-matching induction now proves the converse without counting: if two tagged global words have the same owner-local words, the head occurrence can be moved left through only different-owner entries, after which the proof recurses. Consequently the equivalence closure of the literal adjacent generators is exactly the semantic kernel, and the generated quotient is equivalent to the finite local-word family. This result is now also linearized over the rationals: the kernel of the free-module localization map is exactly the span of the literal two-term adjacent-shuffle rows, and its quotient is linearly equivalent to the free module on local-word families. At the single field-strength-letter level, the ordered-pair space now has an explicit $16 = 6 + 10$ linear decomposition: the six coordinates are the oriented two-planes, while four diagonal and six symmetric coordinates synthesize exactly the span of the literal $F_{ab} + F_{ba}$ rows. Hence the normalization kernel is precisely that row span, the quotient is the six-plane space, and the normalization intertwines the signed $H(4)$ actions. More generally, for every finite family of field slots the product normalization has kernel exactly the span of the literal slotwise antisymmetry rows, with an explicit quotient equivalence to oriented-plane assignments. The antisymmetry and linear shuffle theorems are now assembled while retaining the actual trace layout and then lifted through the dependent dimension-16 union. On that exact carrier, the combined kernel is precisely the span of the already-defined position-indexed antisymmetry and adjacent-shuffle rows; its quotient is the free module on trace layouts, oriented-plane assignments, and owner-local derivative words. The combined map also intertwines the literal signed $H(4)$ actions. That exact fixed-sector target is now explicitly equivalent to the finite labeled local-word carrier: finite derivative functions and owner-local lists are related by inverse maps, the trace permutation and oriented planes are preserved, and the equivalence intertwines field relabeling and the signed $H(4)$ action, including its tensor sign. It therefore descends to an explicit equivalence of the genuine field-label orbit quotients. The actual character is defined independently as the tensor-sign-weighted sum over fixed points of the literal $H(4)$ action on that orbit carrier. A general weighted twisted Burnside theorem, proved by transporter counting and orbit–stabilizer, specializes here to show that the labeled twisted fixed-point numerator is exactly $f!$ times this actual orbit character. The first local factor of the finite cycle theorem is separated semantically: the signed fixed-point sum on one exact local field letter (one oriented plane and an ordered derivative word of length k) is proved to be $\chi_{\Lambda^2 V}(h^r)\chi_V(h^r)^k$ for a relabeling cycle of length r. The exact derivative-degree convolution is now separated as well. Our dependent cycle-seed carrier chooses one exact local letter for each relabeling cycle, lets a cycle of length r act by h^r, and enforces $\sum rk = d$ through exact residual-degree fibers. Its direct signed fixed-point sum is proved equal to the recursive partition derivative character. The literal one-cycle fixed equation is now bridged as well: for every positive r, a family of actual local letters satisfying the one-step relabeling/h equation

is explicitly equivalent to one seed fixed by h^r, and the product of its r one-step tensor signs is proved to be the tensor sign of h^r on that seed. These literal families are now assembled over every positive cycle list in a second dependent carrier: zero-length cycles are uninhabited, exact residual derivative degree is retained, and its complete signed weight sum is proved equal to both the cycle-seed character and the recursive partition character. Our concrete disjoint-cycle carrier takes the recursive sum of the finite blocks displayed by that list and rotates each block by one step. Its literal fixed-decoration subtype is explicitly equivalent to the budget-stable dependent carrier, with exact preservation of the full tensor-sign product. Consequently its signed fixed-decoration character is exactly the recursive partition character. The conjugacy step is now closed as well. The displayed positive partitions through eight fields are proved to be exactly the standard integer partitions; each has a concrete disjoint-cycle representative with the advertised cycle partition. The fiber of the standard partition map on actual field permutations is explicitly equivalent to the corresponding ordinary conjugacy class. The decoration character and centralizer cardinality are constant on that fiber, and the standard class-cardinality times centralizer-cardinality identity cancels the field factorial over $\mathbb{Q}$. Hence the actual signed orbit character equals the recursive cycle-partition character for every $H(4)$ element and every nonempty field sector through eight. The scalar and pseudoscalar values 246,136,792 and 246,121,092 are therefore exact invariant dimensions of this intermediate quotient. They impose none of the remaining physical relations and are neither a physical-basis count nor a compression wall (`HypercubicDimension16CompressionCensus`, `HypercubicDimension16CompressionReplayCertificate`, `HypercubicDimension16CompressionSectorCertificate`, `HypercubicDimension16CompressionSemanticBridge`, `HypercubicDimension16LocalWordCarrier`, `HypercubicDimension16ShuffleQuotient`, `HypercubicDimension16ShuffleLinearQuotient`, `HypercubicDimension16AntisymmetryQuotient`, `HypercubicDimension16ContextualQuotient`, `HypercubicDimension16OrbitCarrierBridge`, `WeightedOrbitBurnside`, `HypercubicDimension16OrbitCharacter`, `HypercubicDimension16LocalLetterCharacter`, `HypercubicDimension16CycleSeedCharacter`, `HypercubicDimension16CycleFamilyCharacter`, `HypercubicDimension16FixedCycleFamilyCarrier`, `HypercubicDimension16CycleBlockDecoration`, `HypercubicDimension16PermutationDecorationTransport`, `HypercubicDimension16PermutationFixedSector`, `HypercubicDimension16PermutationConjugacy`, `HypercubicDimension16PartitionRepresentatives`, `HypercubicDimension16ConjugacyCompression`).

- **The complete raw $H(4)$ character multiplicities are certified.** The standard signed-axis representation, all 384 trace and determinant values, Reynolds projector, and rank-0–16 character averages are checked. At rank 16 the scalar and pseudoscalar tensor multiplicities are 22,379,179 and 22,368,256; multiplication by the exact active-layout count gives the raw invariant-space dimensions displayed above (`HypercubicRawFDCharacterCensus`, `HypercubicRawFDCharacterCertificate`).

- **The Cartan octic chart has exact conditioning.** The 1287 derivative-free monomials split into 17 surviving signed orbits and negative-stabilizer orbits. The orbit indicators, representative duals, reconstruction theorem, and 17×17 identity matrix are certified (`HypercubicDimension16CartanCensus`, `HypercubicDimension16CartanCertificate`, `HypercubicDimension16CartanBasis`).

- **The Wilson coefficient and classical restoration rate are derived on that chart.** Only the pure-octic coordinate is populated, with coefficient $-a^{12}/20160$. The actual Cartan link observable converges to its quadratic $O(4)$-invariant density with the explicit uniform

bound $(5/8)a^4M^4$ (`HypercubicWilsonSymanzikRestoration`).

- **Boundary recombination retains factor one.** The 17 coordinates define a rank-17 degree-zero boundary-cochain quotient; shared-boundary currents cancel exactly. Under the three named analytic bounds, the resulting budget is 5544/5 and the least closing depth is 15 (`HypercubicDimension16BoundarySymanzik`).

- **The frontier is explicit.** The raw character theorem imposes none of the physical relation rows. The derivative-label alpha quotient removes one purely syntactic redundancy. Field-label alpha, adjacent different-owner derivative shuffle, and antisymmetry rows are now equivariant. Innermost Bianchi and on-shell EOM rows are also equivariant, and globally outermost integrated IBP plus traceless one-cycle rows are equivariant. The cross-sector covariant-commutator, traceless anticommutator, and fundamental multitrace rows are now also equivariant. The cycle-index census supplies a literal-group evaluation of its formal expression. Its local-word data, commuting actions, orbit quotient, and sign descent now have a concrete finite realization. The global-to-local derivative map is now an equivariant surjection, and a direct head-matching theorem identifies the full semantic kernel with the equivalence closure of the literal adjacent different-owner generators. On free rational modules, those same literal rows span exactly the kernel of localization. The single-letter field-antisymmetry quotient is now exact and signed $H(4)$-equivariant. Its product lift over every finite family of field slots is also exact: the literal slotwise rows span the full kernel. Its equivariant assembly with the trace context and derivative-shuffle quotient is now exact on the dependent dimension-16 carrier, where the pre-existing position-indexed rows span the full combined kernel. The resulting fixed-sector target is now explicitly equivalent to the finite labeled local-word carrier, compatibly with field relabeling, the signed $H(4)$ action, and tensor sign; this equivalence descends to the genuine field-label orbit quotients. The signed fixed-point character of that quotient is now defined directly, and a weighted Burnside theorem identifies its factorial multiple with the labeled twisted fixed-point sum. The local cycle character and its exact derivative-degree convolution are now realized by a dependent cycle-seed carrier and proved equal to the recursive partition character. On one positive cycle, the literal family fixed by the relabeling/h equation is now explicitly equivalent to the powered fixed seed with exact preservation of the full tensor-sign product. A dependent positive-cycle-list carrier now performs the full assembly with exact residual derivative degree, and its signed sum is exactly the recursive partition character. The carrier is now realized literally as decorations on a recursive disjoint union of finite blocks under blockwise rotation; the explicit equivalence preserves the full signed weight, so its fixed-decoration character is the same partition character. Canonical block representatives now realize every standard integer partition through eight fields, each actual relabeling permutation is conjugate to its selected representative, and the exact partition-fiber equivalence reduces the Burnside sum to ordinary conjugacy classes. Class cardinality cancels centralizer cardinality and the field factorial, proving equality of the actual orbit character and the cycle-index expression in every sector. The standard rows are now assembled into a single policy-indexed linear operator, before and after the exact contextual quotient. Both actions are genuine rational $H(4)$ representations, the operator is equivariant, and its range is invariant. The normalized operator now descends exactly and equivariantly to the genuine field-label orbit carrier: the literal simultaneous-renaming rows are its full kernel and its range is again invariant. The raw rows are now proved source/successor supported, and exact field filters commute through both normalizations, so the descended operator is block upper-bidiagonal by field count. On the genuine eight-field, zero-derivative

orbit module, the signed Reynolds projector and literal character bridge now prove that the scalar invariant target has dimension 11,654. Ordinary two-by-two matrix-index contraction is now proved invariant under simultaneous field relabeling and therefore descends to the genuine orbit carrier. On the implemented pre-quotient sparse module, the singleton traceless rows, adjacent anticommutator rows, and all six rewires of every fundamental trace row are proved to lie in the kernel of this ordinary matrix semantics. The anticommutator proof uses polarized Cayley–Hamilton entrywise; the fundamental proof is the vanishing alternating 3×3 minor with only two matrix-index columns. The checked 98×98 evaluation inverse supplies the matching lower bound, so the complete derivative-free standard trace submodule has finrank 11,556 and its quotient has finrank 98, with explicit exact dual conditioning. This does not close the joint physical quotient: a certified seven-field commutator column has a nonzero eight-field successor component and nonzero Reynolds image, so the field sectors cannot be quotiented independently. Our independent Cartan–Maxwell Hilbert-series socket is now formalized as a bounded fourteen-class character evaluator: its periodic $H(4)$ power traces agree with the literal signed-group data through degree 16, its single-particle resolutions distinguish Bianchi-only and on-shell Maxwell policies, and its cached integer recurrence carries an exact-divisibility ledger for Newton, plethystic, Weyl, and hypercubic averaging. No coefficient row or physical-quotient dimension is claimed from this evaluator here; a bounded certificate replay and a proof that Cartan specialization descends through the full joint relation quotient are still absent. The coordinate-free joint physical relation submodule is now defined before any row ordering, and its off-shell-to-on-shell inclusion is exact. The 11,556 transferred trace rows, one seven-to-eight commutator row, and one four-to-five-field commutator row give a certified joint relation-rank lower bound of 11,558 for either policy; the last is separated by an exact field-five Reynolds coordinate $-1/24$. The bound also holds for the Leibniz-corrected joint submodule, whose commutator rows are field identities. The remaining finite problem is the complete joint rank and conditioning after Bianchi, EOM, IBP, commutator, and trace rows are assembled across all sectors. The quantitative restoration theorem is restricted to bounded constant Cartan curvature and is not a uniform interacting Symanzik theorem. Constructive Yang–Mills additionally requires a continuum measure, OS reconstruction, and a positive spectral gap.

F Trace-order and coupled-relation repairs

The next finite analysis makes the repaired finite relation target coordinate-free. For either relation policy, the invariant physical relation submodule $\mathcal{J}_p$ is defined as the preimage of the full physical relation range inside the signed $H(4)$-invariant orbit space. The exact off-shell-to-on-shell inclusion is proved. A family of 11,556 transferred field-eight trace relations, an independent seven-to-eight commutator row, and an explicit four-to-five-field commutator row with Reynolds coordinate $-1/24$ give the certified bound

$$11558 \leq \dim_{\mathbb{Q}} \mathcal{J}_p.$$

This is a lower bound, not a complete joint rank, quotient dimension, or conditioning certificate. The commutator row used there is a field identity only when its field carries no outer derivative; the same bound holds for the Leibniz-corrected submodule (§C, the correction to the covariant-derivative commutator). Our finite derivative-alpha-reduced $F, D/\mathrm{IBP}$ relation cochains now map by the physical relation operator followed by signed $H(4)$ Reynolds averaging onto this same J_p. The map is $H(4)$-equivariant, surjective for both policies, and has range equal to the whole J_p; shared off-shell cochains retain their ambient invariant semantics on shell. This identifies the

repaired finite relation object. It does not supply the still-missing full noncommutative Wilson-functional coordinates or analytic dual jets. A further exact comparison fixes why the existing Cartan chart cannot be silently reused for that task: the two literal eight-field endpoints of a certified incoming covariant-commutator column have the same unordered Cartan plane decoration but different trace orders. Our raw Cartan restriction of the actual Wilson eighth-jet coefficient kills their signed difference, while its ordinary derivative-free trace-quotient class is certified nonzero. Independently, that trace quotient has dimension 98, so no map from it into a 17-coordinate Cartan chart can be injective. More sharply, no ordinary-trace-quotient coordinate which agrees with the actual Cartan eighth-jet scalar on every raw labelled input can even detect that certified incoming trace-order class; an injective realization of this scalar coordinate is therefore formally uninhabited. This is not a no-go for the Wilson route. It specifies the repair: a full Wilson coordinate system must retain noncommutative trace order and the coupled lower-field commutator component in additional coordinates (`HypercubicDimension16WilsonCartanJointMismatch`, `HypercubicDimension16WilsonCartanTraceOrderFaithfulnessNoGo`).

Formal repair. The first finite repair requirement is now exact. On the concrete seven/eight-field commutator quotient, the canonical eight-field trace coordinate cannot descend by itself. More sharply, every descended trace-order correction has a forced nonzero value on the seven-field component of the coupled row. The explicit compensator realizes that value; it kills the complete projected physical commutator for either relation policy while still detecting its isolated eight-field trace component. We package this rational quotient coordinate beside the exact Cartan Wilson coefficient packet, not as an identification of the two and not as a completed analytic noncommutative Wilson functional (`HypercubicDimension16WilsonTraceOrderRepair`).

Formal repair. The trace-order layer is now fully coordinatized on this bounded quotient. The existing 98 exact ordinary trace duals assemble into an injective packet with a checked synthesis map recovering every ordinary trace class. Composed with the forced seven-field compensator, this packet kills the full projected physical commutator for either relation policy while retaining its isolated eight-field component as the exact nonzero two-pivot trace vector. This makes a finite noncommutative trace-order coordinate layer available beside the Cartan packet; it does not identify that rational layer with an analytic Wilson functional, establish descent through the full relation range, or supply a conditioning estimate for the added seven-field compensator (`HypercubicDimension16WilsonTraceOrderCoordinateCompletion`).

Formal repair. The first signed-hypercubic extension is then tested rather than assumed. A one-coordinate extension kills the commutator and derivative-free trace rows but has an explicit nonzero value on a physical IBP row. A second field-relabel-invariant split-derivative/plane coordinate supplies exactly the missing independent value. The resulting two-coordinate signed-$H(4)$-invariant correction kills the concrete commutator, the displayed IBP row, and every lifted derivative-free eight-field trace generator. A further exact test constructs a genuine three-distinct-axis innermost Bianchi row; both coordinates have zero signed weights on its three normalized summands, so the correction kills the row under either policy and its Reynolds-averaged invariant relation member. An explicit on-shell EOM row has four contracted-axis terms; both coordinates have zero signed weights on each term, while exact normalization removes the diagonal term and leaves orientations $1, 1, -1$. Thus the correction also kills that physical row and its Reynolds-averaged on-shell relation member.

Formal repair. The next genuine IBP row is not silently absorbed. It has a different split derivative pattern, and the two-coordinate correction has the exact nonzero value $(1/2)$ times the incoming trace class on it. A third field-relabel-invariant plane/derivative coordinate has calibrated value 1 on this row and signed weight 0 on the already checked commutator, first IBP, Bianchi, and EOM rows. Subtracting one half of its trace contribution therefore kills the new physical

row and its Reynolds-averaged invariant relation member while retaining those earlier physical and invariant checks. This is a finite repair family: it neither establishes annihilation of the full Bianchi/EOM/IBP/trace relation range nor constructs noncommutative Wilson-functional analytic coordinates (`HypercubicDimension16WilsonTraceOrderInvariantRepair`, `HypercubicDimension16WilsonTraceOrderInvariantBianchi`, `HypercubicDimension16WilsonTraceOrderInvariantEOM`, `HypercubicDimension16WilsonTraceOrderInvariantThreeCoordinateRepair`).

Formal repair. The next canonical trace family supplies an exact boundary test rather than an implicit closure claim. On the same decorated seven-field carrier, the polarized SU(2) trace-anticommutator row has source, adjacent-swap, and split-trace terms. The three existing coordinates see the same field and derivative decorations in all three terms: the first has total value $1/2$, while the two plane/derivative coordinates vanish. The three-coordinate correction therefore has the exact nonzero value $(1/2)$ times the incoming trace class, both on the physical row and on its Reynolds-averaged member of J_p, for either policy. Thus it cannot descend through the complete relation range (`HypercubicDimension16WilsonTraceOrderTraceAnticommutatorObstruction`).

Formal repair. That failure has a finite, trace-topological repair. We gate the derivative-axis character by the relabel-invariant condition that the trace permutation has exactly one two-cycle. The source and swap have cycle type $\{7\}$, while the split term has type $\{2,5\}$; their exact signed weights are respectively 0, 0, and 32. The resulting fourth scalar is -1 on the full trace-anticommutator row, so adding one half of its incoming trace-class contribution cancels the preceding $(1/2)$ residual. Kernel replay also gives zero on the displayed commutator, both explicit IBP rows, Bianchi, on-shell EOM, and every lifted derivative-free eight-field trace row, before and after Reynolds averaging. This repairs the displayed finite family only: it does not establish descent through all of J_p, a complete relation census, or a noncommutative Wilson-functional coordinate system (`HypercubicDimension16WilsonTraceOrderTraceTopologyRepair`).

Formal repair. The canonical fundamental three-cut identity tests that repair on a distinct six-term trace wiring. Its six summands have cycle types $\{7\},\{6\},\{2,5\},\{6\},\{7\},\{5\}$. The four-coordinate functional therefore has exact value $-\frac{1}{2}$ times the incoming trace class, on both the physical row and its Reynolds member. A fifth scalar gates the same derivative-axis character by the presence of one six-cycle. Its exact signed weights on those six summands are $0, 32, 0, 32, 0, 0$, giving normalized value -2; adding $-\frac{1}{4}$ of its incoming trace contribution cancels the residual. The new scalar has zero value on the displayed trace-anticommutator row, so the two named canonical families remain killed before and after averaging. This is again a finite named-family repair, not a proof of descent through all of J_p or a complete Wilson-functional coordinate construction (`HypercubicDimension16WilsonTraceOrderFundamentalThreeCutRepair`).

Formal repair. The two individual gate repairs admit a more constrained finite continuation. We enumerate all 210 ordered choices of three distinct cut labels on the same seven-cycle, form the alternating six-term trace-cycle profile, and find four topology equations. Together with the polarized trace-anticommutator equation, they uniquely force the coefficients

$$(c_2, c_3, c_4, c_5, c_6) = \left(\tfrac{6}{5}, \tfrac{13}{10}, -1, -\tfrac{7}{10}, -\tfrac{3}{5}\right)$$

for the gates recording one cycle of lengths two through six. Kernel replay checks that this full rational profile vanishes for every one of the 210 ordered placements. We lift it to a signed-hypercubic coordinate and verify that it kills the two displayed canonical physical rows and their Reynolds-averaged members. Each certified placement is now mapped to a distinct actual fundamental-trace site and policy-indexed physical relation generator; its finitely supported source cochains embed injectively in our F, D/IBP cochain space and map by signed Reynolds semantics to the coordinate-free joint relation submodule. The canonical label recovers the previously displayed fundamental

row. A separate symbolic field-sector transport proves that the whole finitely supported physical three-cut family has zero exact field-eight invariant projection for either relation policy: all six trace-rewire terms remain seven-field. Our physical trace-cycle bridge next factors every rewire's signed Reynolds weight through the same profile gate, replays the six-gate cancellation on all 210 actual labels, and proves that the invariant trace-cycle-profile coordinate annihilates every finitely supported cochain in this physical family under either policy. The three earlier coordinate weights are trace-order blind: rewiring changes the trace but not the decorated local word, so the six signed weights agree and their alternating coefficients cancel identically. We transport that identity through the actual physical relation operator and prove that the separate three-coordinate correction also annihilates every cochain in this family. Thus the full current cycle-profile correction is zero on this realized physical family under either policy. This closes the field-eight sector and all current correction coordinates for this family only; it does not identify every physical source generator, establish descent through all of J_p, give a complete relation census or conditioning certificate, or construct noncommutative Wilson-functional coordinates (`HypercubicDimension16FundamentalTraceCycleProfile`, `HypercubicDimension16WilsonTraceOrderCycleProfileRepair`, `HypercubicDimension16FundamentalTracePhysicalFamily`, `HypercubicDimension16FundamentalTracePhysicalFieldEight`, `HypercubicDimension16FundamentalTracePhysicalCycleProfile`, `HypercubicDimension16FundamentalTracePhysicalThreeCoordinate`).

Formal repair. The polarized trace-anticommutator test is now extended from its displayed placement to all seven adjacent placements on the same actual physical seven-cycle. Each physical row has the unrewired, adjacent-swap, and split-trace terms with one common field/derivative decoration. The three trace-order-blind coordinates consequently retain the common $\frac{1}{2}$ incoming trace-class residual. A kernel-reduced finite replay of the seven trace topologies gives the common profile value $-\frac{1}{2}$, so the profile contribution cancels that residual at every placement. The seven basis rows embed injectively into the policy-indexed $F, D/\mathrm{IBP}$ source and their finitely supported cochains map by signed Reynolds semantics into the coordinate-free joint relation submodule; the full current cycle-profile correction vanishes on this whole finite family under both policies. This is a second realized trace-family block, not a census of trace-anticommutator sites on arbitrary carriers or a descent theorem for all of J_p (`HypercubicDimension16TraceAnticommutatorPhysicalFamily`).

Formal repair. The full profile correction is also tested against the five displayed non-trace calibration rows of the physical relation operator: the covariant commutator, the original and second integration-by-parts rows, Bianchi, and the on-shell equation-of-motion row. For every normalized seven-field summand, the signed $H(4)$ orbit sum of the new profile multiplicity is replayed exactly in the kernel; each relevant sum is zero. Linearity and the already-identified physical rows then show that the full correction kills both the physical rows and their Reynolds-averaged invariant representatives. Those representatives are genuine members of the coordinate-free joint relation submodule. An arbitrary-carrier bridge then identifies every physical source-eight traceless, polarized-anticommutator, and fundamental three-cut row with the derivative-free field-eight schema. The derivative-bearing site types are empty at that source count. Thus the full correction annihilates every source-eight physical relation column, not only a chosen lifted family. Our source-six curvature-pair bridge next handles the sole cross-sector covariant-commutator branch: its two seven-field insertions have the same three decoration coordinates and conjugate trace permutations, hence the same signed-Reynolds profile coordinate, so their opposite row signs cancel. Combined with the field-support theorem below six, the full correction now annihilates every arbitrary physical relation column outside the source-seven band. This is not a source-seven census, a complete physical relation census, a proof that these rows span J_p, or descent through all of J_p (`HypercubicDimension16WilsonTraceOrderCycleProfileSourceSix`, `HypercubicDimension1`

`6FieldEightTracePhysicalCompleteness`). The same holds for the Leibniz-corrected operator of Section C: there a six-field commutator row has one conjugate insertion pair for each subset of its outer word, and each pair cancels (`HypercubicDimension16LeibnizSourceSix`).

Formal repair. The remaining source-seven boundary is now narrowed by an exact counterexample to the present *one-class* repair, rather than left as an undifferentiated census task. We construct a second physical covariant-commutator carrier whose two curvature insertions have plane profile $(0,0,1,0,0,0,0,1)$ and different trace orders. Its signed eight-field pair has a kernel-checked value $-4/3$ under an ordinary derivative-free trace quotient functional. At the same evaluation assignment, the original incoming class used by each of the existing seven-field compensator coordinates has value zero. Consequently the functional evaluates the full current cycle-profile correction on this actual row to $4/3$, so that correction is nonzero under both physical relation policies. This refutes only the current correction family, whose compensators all take values in the single original incoming trace class. We then construct a bounded two-class repair of these two actual rows. Two field-relabel-invariant signed-$H(4)$ selectors count a whole two-derivative word together with the plane on a distinct field. Their exact values on (incoming, secondary) are $(1,0)$ and $(0,1)$, respectively. Rather than substitute that standalone map for the prior correction, we replay the new secondary selector on the already checked first-IBP, Bianchi, on-shell EOM, fundamental three-cut, polarized trace, and lifted field-eight rows; every listed value is zero. The genuine second-IBP row instead has normalized selector value 1. Let R be the exact (and certified nonzero) residual of the prior correction on the secondary row. Our two-class augmentation is the prior correction minus the secondary selector times R; on the second-IBP row it has exact value $-R$, so this is a machine-checked incompatibility rather than an omitted check. We then use the independent third split-derivative/plane coordinate already calibrated by the earlier three-coordinate construction. Its signed sums vanish on both displayed source-seven commutators and its normalized second-IBP value is 1. Adding this third coordinate times R produces our three-class correction. It cancels both commutators and the named first-IBP, second-IBP, Bianchi, on-shell EOM, fundamental, polarized-trace, and lifted field-eight rows under either relation policy. A further actual source-seven singleton-trace row then refutes this three-class architecture under both policies. It shares the incoming two-derivative word but has one genuine singleton trace. Its exact signed weights for the original derivative, IBP, third, and cycle-profile coordinates are 32, 0, 0, and $-96/5$, respectively; both plane-profile selectors vanish. Hence the full three-class correction has exact value $(-1/10)$ times the certified nonzero incoming trace class. More generally, every correction obtained from the cycle-profile term by assigning arbitrary quotient values to these same three selectors has that value on this row, so no adjustment of those three added selector values, with the cycle-profile base held fixed, can descend through the physical relation submodule. The next finite task is to solve the exact compatibility system after also allowing the existing base scalar coordinates to vary. This is a machine-checked mismatch, not a complete source-seven census, joint rank or conditioning calculation, or Wilson-functional coordinate system (`HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenMismatch`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenCorrectionRefutation`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenTwoClassRepair`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenTwoClassCompatibility`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenThreeClassRepair`, `HypercubicDimension16WilsonTraceOrderCycleProfileSourceSevenThreeClassTracelessRefutation`). The first exact reparameterization test is now positive. Without adding a coordinate, we form the signed-$H(4)$-invariant scalar direction $-A+\frac{1}{2}B-\frac{1}{2}C-P+I+S$, where A,B,C,P are the established derivative, two IBP, and cycle-profile coordinates and I,S are the two plane-profile selectors. Multiplying that direction by the already certified incoming trace class and adding it to

our three-class correction contributes exactly $+\frac{1}{10}$ of that class on the singleton row. It therefore cancels the previous $-\frac{1}{10}$ value, while its exact values on both explicit source-seven commutators are zero. Further kernel-reduced signed-orbit replays give zero on the displayed first-IBP row under the off-shell policy, on the second-IBP and Bianchi rows under either policy, on the displayed on-shell EOM row, on the canonical fundamental three-cut row, and on the polarized trace-anticommutator row. This is a checked repair on nine actual named rows, but it does not descend. A second actual singleton-trace relation retains the independent derivative geometry of the second-IBP carrier while making field 0 a genuine one-letter trace. Its six signed weights are

$$[32, 0, 0, -96/5, 0, 16].$$

Thus its third-coordinate value is 0 and its secondary plane-profile value is 1. The already certified quotient functional annihilates the literal incoming class but evaluates the stored secondary residual as $4/3$; it therefore evaluates the complete six-scalar repair as $-4/3$ under either physical policy. This is a machine-checked refutation of descent for that specific scalar family, not a refutation of an enlarged coordinate family or of a semantics-preserving canonical (F, D) bridge (`HypercubicDimension16WilsonTraceOrderExistingScalarParameterRepairTracelessRefutation`). The functional statement is stronger than the displayed six-coefficient calculation: every map obtained from the fixed three-class correction by adding an arbitrary scalar-valued functional times the literal incoming class has the same value $-4/3$ on this witness, because the separating functional annihilates that entire literal direction. Thus no scalar-literal extension can descend; a repair must change a non-literal quotient direction or the three-class base itself (`HypercubicDimension16WilsonTraceOrderThreeClassScalarExtensionRefutation`). The most direct prospective extra coordinate has also been screened exactly: gating the derivative-axis character by one genuine fixed trace point detects the singleton row with numerator 32, but has complete fundamental-three-cut numerator -64, hence singleton-normalized value -2 on a row the current correction annihilates. This gate alone therefore cannot be appended with a nonzero target; any such topology extension needs a companion cancellation direction (`HypercubicDimension16WilsonTraceOrderFixedPointTopologyScreen`).